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"Vegetation and Energy"
"CGLOPS-1"
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VALIDATION REPORT
UPDATE 2024

SENTINEL-3 TOP OF CANOPY (TOC) REFLECTANCE
VERSION 1

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List of Acronyms

AATSR	Advanced Along-Track Scanning Radiometer
AD	Applicable Document
AIR	Aerospace Information Research Institute
APR	Annual Performance Report
APU	Accuracy, Precision and Uncertainty
ATBD	Algorithm Theoretical Basis Document
B	Mean Bias
BSCN	Baoutou Sand ChiNa
BTCN	BaouTou ChiNa
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land OPerationS
CLMS	Copernicus Land Monitoring Service
CNES	Centre National D'Etudes Spatiales (French Space Agency)
DEGE	DEmmin GErmany
EOLAB	Earth Observation LABoratory
ESA	European Space Agency
GHNA	Gobabeb HYPERNETS NAmibia
GONA	GObabeb NAmibia
HBK	HandBooK
HYPERNETS	HYPERspectral radiometer integrated in automated NETworkS
JRC	Joint Research Centre
LCFR	La Crau FRance
LPV	Land Product Validation subgroup of CEOS
L2A	Level 2A
MAD	Median Absolute Deviation
MAR	Major Axis Regression
MD	Median Deviation
MERIS	Medium Resolution Imaging Spectrometer
MODIS	Moderate Resolution Imaging Spectroradiometer
N	Number of samples
NIR	Near InfraRed
OLCI	Ocean and Land Colour Instrument
OMPC	Optical Mission Performance Centre
PN	Product Notice
PUM	Product User Manual
QAR	Quality Assessment Report
R	Correlation coefficient
RadCalNet	Radiometric Calibration Network

R_{CV}	Relative Coefficient of Variation
RMSD	Root Mean Square Deviation
R_{SE}	Scale Requirement Index
R_{ST}	Relative STrength of the spatial correlation
R_{SV}	Relative proportion of the Structural Variation
RVUS	Railroad Valley USA
RAWscore	first-order score or spatial representativeness
S3	Sentinel-3
SDR	Surface Directional Reflectance
SI	Système International (d'unités)
SLSTR	Sea and Land Surface Temperature Radiometer
SQAD	System Quality Assurance Document
SRF	Spectral Response Function
STscore	Standard score for spatial representativeness
STD	Standard deviation
SWIR	Short-wave infrared
SYN	SYNergy
TG	Technical Guide
TOA	Top-of-Atmosphere
TOC	Top-of-Canopy
UTC	Universal Time Coordinated
VIIRS	Visible Infrared Imaging Radiometer Suite
VNIR	Visible and Near InfraRed
VZA	Viewing Zenith Angle
WGCV	Working Group on Calibration and Validation
WWUK	Wytham Woods United Kingdom

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

This report is the 2024 update of the quality assessment report of the Sentinel-3 Top-Of-Canopy reflectances (TOC) version 1. It presents the results of the comparison of the TOC reflectances version 1, generated from the Ocean and Land Colour Instrument (OLCI) and Sea and Land Surface Temperature Radiometer (SLSTR) observations, with some in-situ reference datasets. In particular, ground data from Committee on Earth Observation Satellites (CEOS) Radiometric Calibration Network (RadCalNet) is used as main reference, and complemented with datasets from hyperspectral radiometers integrated in automated networks (HYPERNETS, 2021-2023 period due to their availability). ESA L2 Synergy (SYN) products are also compared with ground measurements for benchmarking.

The direct validation of Sentinel-3 TOC reflectances V1 2024 retrievals close to nadir (VZA < 10º) with CEOS RadCalNet multi-temporal data from 4 sites (desert and sparse vegetation) shows similar results than previous periods and cumulative whole period (July 2018 – December 2024) in terms of accuracy (mean bias, median deviation). Better results are found for 2024 products compared with the whole cumulative and previous periods in terms of overall uncertainty. The accuracy is typically around ±2% for most OLCI channels in 2024 while the overall uncertainty is typically better than 6%. Positive bias of 7%, 3% and 2% is found for SLSTR S1, S2 and S3 SLSTR channels and negative bias of -9% and -12% for S5 and S6. Satisfactory conformity testing results are found for 2024 products when no uncertainties are considered (i.e., simple acceptance), with typically around 40%-50% of pixels within goal requirement and 55%-70% considering threshold level. However, mostly inconclusive results are found when satellite product and RadCalNet measurements uncertainties are considered for conformity testing (i.e., guarded acceptance), with around half of the cases inconclusively non-conforming and the other half inconclusively conforming regarding the threshold level. The temporal trajectories of CLMS TOC reflectances V1 from both OLCI and SLSTR sensors show similar temporal trends than RadCalNet measurements. Temporal trends are also consistent between different years and platforms (i.e., Sentinel-3A and -3B), and the bias over GONA pseudo-invariant site is reduced from July 2022 onwards due to the correction of the geolocation issue in CLMS processing chain. CLMS TOC reflectances V1 show better results than ESA L2 SYN, except for SLSTR SWIR channels (as expected due to incorporation of correction coefficients in the ESA L2 SYN processing).

The accuracy of Sentinel-3 TOC reflectance V1 retrievals compared with HYPERNETS multi-temporal and multi-angular data from 3 sites (desert, agricultural and deciduous broadleaf forest) ranges from 4.4% up to 9.5% for OLCI channels. Worse accuracy is found for SLSTR (MD of 14.3% and 9.4% for S1 and S2, MD of 3.6% for S3 MD of -10.3% for S5). CLMS performance is better to

that found for ESA L2 SYN for most APU metrics and channels in the comparison with HYPERNETS.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVE

The operational quality monitoring encompasses an annual scientific validation of the most recent products in order to check the stability of the product quality as described in the System Quality Assurance Document [CGLOPS1_SQAD]. The objective of this annual scientific validation is to update the direct validation results of the version 1 of TOC reflectances 300m products (referred hereafter as TOC V1) generated from data of the Ocean and Land Colour Instrument (OLCI) and Sea and Land Surface Temperature Radiometer (SLSTR) onboard Sentinel-3 (S3). For the year 2024, TOC reflectances v1.1.0 were produced and are evaluated. Version 1.1.0 corrects, from 1st July 2022 onwards, the geolocation issue of the previous v1.0.1 detected in the original quality assessment report [CGLOPS1_QAR_S3-TOCr300m-v1].

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 describes the methodology for operational quality monitoring
- Chapter 3 presents the results of the analysis
- Chapter 4 summarizes the main conclusions and recommendations

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_SQAD	System Quality Assurance Document of the Copernicus Global Land Operations "Vegetation & Energy"
CGLOPS1_QAR_S3-TOCr300m-v1	Quality assessment report of Sentinel-3 OLCI and SLSTR TOC reflectances version 1.

Available at https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document ID	Descriptor
ISO-10576:2022	Statistical methods — Guidelines for the evaluation of conformity with specified requirements

Available at https://www.iso.org/es/contents/data/standard/07/88/78878.html?browse=ics

S3-OLCI TG Sentinel-3 OLCI instrument Technical Guide.

Available at https://sentiwiki.copernicus.eu/web/s3-olci-instrument

S3-SLSTR TG Sentinel-3 SLSTR instrument Technical Guide.

Available at https://sentiwiki.copernicus.eu/web/s3-slstr-instrument

L2-SYN-SDR-processing ESA L2 SYN Surface Directional Reflectance processing

Available at https://sentiwiki.copernicus.eu/web/synergy-products#S3-Synergy-Products-L2-SYN-SDR-processing

S3-L2-SD-03-S02-ATBD SYN Algorithm Theoretical Basis Document

Available at https://sentiwiki.copernicus.eu/__attachments/1672112/S3-L2-SD-03-S02-ATBD_-Sentinel-3 Synergy Level-2 and Level-3 ATBD 2017 - 2.5.pdf

L2-SYN-processing-baseline Sentinel-3 L2 SYN processing baseline

Available at https://sentiwiki.copernicus.eu/web/synergy-processing#Processing-Baseline

S3.PN-SYN-L2.10 Sentinel-3 Product Notice- SYNergy. Issue/Rev Date 29/02/2024.

Available at https://sentinel.esa.int/documents/d/sentinel/s3-pn-syn-l2-10-sentinel-3-synergy-product-notice-level-2-2024-1-0

OMPC.ACR.HBK.003 Copernicus Sentinel-3 SYN Land User Handbook

Available at https://sentinel.esa.int/documents/247904/4598110/Sentinel-3-Synergy-Land-Handbook.pdf

OMPC.ACR.APR.004 Sentinel-3 Optical Annual Performance Report – Year 2023

Available at https://sentiwiki.copernicus.eu/__attachments/1681931/OMPC.ACR.APR.004_-_S3_Optical_Annual_Performance_Report_2023_-_1.1.pdf

2 OPERATIONAL QUALITY MONITORING METHODS

2.1 DIRECT VALIDATION AND CONFORMITY TESTING

Accuracy, Precision and Uncertainty (APU) are evaluated by several metrics (Table 1) reporting the goodness of fit between the satellite products and the corresponding reference dataset.

For the study period (from 1st January 2024 to 31st December 2024), the temporal realism and the error evaluation of CLMS Sentinel-3 TOC V1 products are analyzed as compared to Radiometric Calibration Network (RadCalNet) and HYPERNETS multi-temporal observations. Additionally, the Sentinel-3 SYN Level-2 product (ESA L2 SYN) from European Space Agency (ESA) is also compared with ground data, for benchmarking.

Table 1: Validation metrics for product validation. 'x' refers to reference products, and 'y' to product under evaluation

Statistics	Comment
N	Number of samples. Indicative of the power of the validation
B	Mean Bias. Difference between average values of x and y. Indicative of accuracy and offset. Bias (%) is the relative mean bias between the average of x and y.
MD	Median deviation between x and y. CEOS LPV best practice reporting the accuracy. MD (%) is the relative MD between the average of x and y.
STD	Standard deviation of the pair differences. Indicates precision. STD (%) is the relative STD between the average of x and y.
MAD	Median absolute deviation between x and y. CEOS LPV best practice reporting the precision. MAD (%) is the relative MAD between the average of x and y.
RMSD	The Root Mean Square Deviation is the square root of the average of squared errors between x and y. CEOS LPV best practice reporting uncertainty. RMSD (%) is the relative RMSD between the average of x and y.
MAR	Slope and offset of the Major Axis Regression linear fit. Indicates some possible bias.
R	Correlation coefficient. Indicates descriptive power of the linear accuracy test. Pearson coefficient is used.

The final objective of the quality assessment analysis is to verify how much the products are compliant with the user requirements. Conformity testing is the process that determines if the estimated quantities are within the range of tolerable values or not (Widlowski, 2015). CLMS Sentinel-3 TOC V1 products are evaluated with respect to user requirements in terms of uncertainty

according to applicable document [AD1]. In particular, both goal (<3%, 2-sigma) and threshold (<5%, 2-sigma) requirements on uncertainty are considered.

Up to now, in most studies, satellite and references uncertainties have not been considering in the conformity testing. In those cases, the data points which are within the tolerance intervals (requirements) are considered conform. This decision rule is known in metrology as "**shared risk**" or "**simple acceptance**" conformity testing. Even if the shared risk approach can provide a first assessment of the conformity with requirements, it is necessary to consider both, product and references uncertainties, for a better understanding of the compliance of requirements (Camacho et al., 2024). In this case, a product value is considered conform when its coverage interval at a coverage factor (k) is fully within a tolerance interval (requirements). The use of uncertainty information reduces the risk of false acceptance or false rejection of data. For that reason, this decision rule is known in metrology as "**guard banding**" or "**guarded acceptance**" [ISO-10576:2022]. Coverage factor k=2 (95.4% confidence) is used, which corresponds to expanded uncertainties (2-sigma) as defined in applicable document [AD1].

In guard banding or guarded acceptance, the apparent error (ε_i) and its coverage interval ($\varepsilon_i \pm u(\varepsilon_i)$) are calculated as follows:

$$\varepsilon_i = y_i - x_i \quad \text{Eq. 1}$$

$$u(\varepsilon_i) = \sqrt{u(y_i)^2 + u(x_i)^2} \quad \text{Eq. 2}$$

where y_i is the product under evaluation and x_i the reference values.

Then, according to a maximum permissible error (Δ) or user's requirement, each product sample falls in one the following category (k=coverage factor), as illustrated in Figure 1:

- Conform or guarded acceptance: If the absolute value of the apparent error and its coverage interval are lower or equal than the maximum permissible error.

$$|\varepsilon_i \pm ku(\varepsilon_i)| \leq \Delta \quad \text{Eq. 3}$$

- Non-conform or guarded rejection: If the absolute value of the apparent error and its coverage interval are greater than the maximum permissible error.

$$|\varepsilon_i| > \Delta \;\&\; |\varepsilon_i \pm ku(\varepsilon_i)| > \Delta \quad \text{Eq. 4}$$

- Inconclusive conform: If the apparent error absolute value is lower or equal than the maximum permissible error, but the coverage interval of the error is greater than the maximum permissible error.

$$|\varepsilon_i| \leq \Delta \;\&\; (|\varepsilon_i + ku(\varepsilon_i)| \text{ or } |\varepsilon_i - ku(\varepsilon_i)|) > \Delta \quad \text{Eq. 5}$$

- Inconclusive non-conform: If the apparent error absolute value is greater than the maximum permissible error, but the coverage interval of the error is lower or equal than the maximum permissible error.

$$|\varepsilon_i| > \Delta \;\&\; (|\varepsilon_i + \mathrm{ku}(\varepsilon_i)| \text{ or } |\varepsilon_i - \mathrm{ku}(\varepsilon_i)|) \leq \Delta \qquad \text{Eq. 6}$$

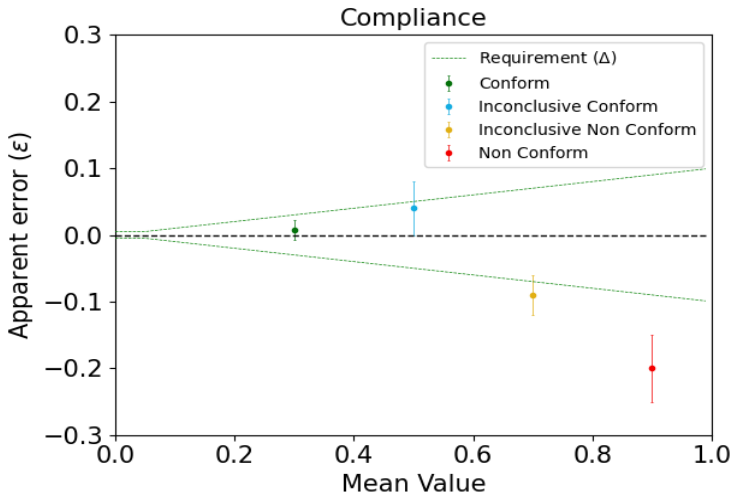


Figure 1. Overview of the four possibilities in guarded acceptance conformity testing. Green dashed lines represent the maximum permissible error.

2.2 IN-SITU REFERENCE DATASET

2.2.1 Radiometric Calibration Network (RadCalNet)

RadCalNet (Bouvet et al., 2019) is an initiative of the Committee on Earth Observation Satellites (CEOS) Working Group on Calibration and Validation (WGCV). RadCalNet is a network of sites that can be used to compare different satellite sensors to each other and to a common reference. For each site, measurements of nadir-viewing TOC reflectance along with atmospheric parameters (such as surface pressure, columnar water vapor, columnar ozone, aerosol optical thickness, and the Angstrom coefficient to define aerosol sizes) are provided. Simulation to TOA reflectance is performed through the central RadCalNet Processing for all sites, ensuring that the in-situ data from those sites are SI-traceable (SI – from Système International (d'unités)). TOA reflectance and TOC reflectance data are provided for a nadir view at 30 min intervals from 9 am to 3 pm local standard time for a given site, along with their associated uncertainties, to users through the RadCalNet portal[1].

[1] https://www.radcalnet.org (RadCalNet data access on 14th January 2025)

NOTE: RadCalNet detected an instrument issue for GONA site from 13th August 2023 to May 2024, resulting in data gap or higher RadCalNet uncertainty than usual (see RadCalNet forum: https://forum.radcalnet.org/).

As we detected several outliers close to the instrument issue (before and after), GONA data from 1st July 2023 to 30th June 2024 was removed from our study.

Two assumptions are considered to deal with differences in data acquisition between Sentinel-3 TOC reflectances and RadCalNet ground reference measurements:

- Similar illumination and viewing configurations are considered. For that, Sentinel-3 observations with viewing zenith angle higher than 10º are discarded, as RadCalNet are measured at nadir viewing angle. Same dates and acquisition times were compared.
- The spectral range of RadCalNet measurements is from 380 nm to 2500 nm (except for BSCN site, with maximum wavelength of 1000 nm), with 10 nm of spectral sampling interval. These narrow-band measurements are made comparable with Sentinel-3 channels using the Spectral Response Function (SRF) coefficients of Sentinel-3 [S3-OLCI TG, S3-SLSTR TG].

Table 2 summarizes the main information of the 4 RadCalNet sites used in this study. For each site, the spatial representativeness of the measurement was evaluated. Note that RadCalNet also provides data for a fifth site (Baotou - BTCN), but was discarded as it relies on a very small artificial surface.

Table 2: Characteristics of the four RadCalNet sites considered in this study.

ID Site	Name	Country	Latitude	Longitude	Land Cover	Institution
GONA	Gobabeb	Namibia	-23.59999°N	15.119215° E	desert	ESA/CNES
LCFR	La Crau	France	43.56° N	4.86° E	sparse vegetation	CNES
RVUS	Railroad Valley Playa	USA	38.497° N	-115.69° E	desert	University of Arizona
BSCN	Baotou Sand	China	40.8658° N	109.6155° E	sand desert	AIR

The reference reflectance measured from towers cover a circular footprint that varies with the tower height. It is unlikely that the footprint of the ground measurements exactly matches the satellite pixel sizes. Then, the spatial representativeness of the tower-based measurements should be evaluated to minimize the issues associated with spatial representativeness in the point-to-pixel comparison (Román et al., 2010, 2009).

We adopted CEOS LPV recommendations and albedo validation protocols (Wang et al., 2019) for the analysis of the spatial representativeness of tower based measurements. The semi-variogram (Carroll and Cressie, 1996; Hohn, 1991) is one of the most efficient tools for describing spatial representativeness. Semi-variograms are computed for the stations under evaluation using Sentinel-2 surface reflectances at a spatial resolution of 10 m near-nadir. The spatial attributes (e.g., the range, sill and nugget) are computed by fitting the variogram estimator to an isotropic spherical model (Matheron, 1963), and they can reveal the spatial variability of land surfaces and the scaling effects associated with remotely sensed data (Román et al., 2010, 2009; Wang et al., 2017, 2014, 2012; Woodcock et al., 1988a, 1988b).

The methodology adopted in this study for the estimation of the geostatistical indexes is based on the comparison of the variogram model parameters retrieved at different spatial resolutions (i.e., from 300 m to 500 m squared subsets, to allow for errors in geolocation for 300 m resolution). Four different geostatistical attributes are used (Cescatti et al., 2012; Román et al., 2010): the relative coefficient of variation (R_{CV}), the scale requirement index (R_{SE}), the relative strength of the spatial correlation (R_{ST}) and the relative proportion of the structural variation (R_{SV}). The four geostatistical attributes can be combined in a compact metric (ST_{score} (Román et al., 2010, 2009), see Eq. 1), which represents a spatial representativeness score using R_{SE} as the primary marker and the others like secondary weights. When the spherical variogram model does not provide a good fit to the variogram estimator, another indicator (RAW_{score} (Román et al., 2010, 2009), see Eq. 2) could be used to provide a spatial representativeness score, which is only based on the R_{CV}.

$$ST_{score} = \left(\frac{|R_{CV}|+|R_{ST}|+|R_{SV}|}{3} + R_{SE}\right)^{-1} \quad \text{Eq. 1}$$

$$RAW_{score} = |2\ R_{CV}|^{-1} \quad \text{Eq. 2}$$

Both scores are directly proportional to the site representativeness. In case of desert sites (GONA, RVUS and BSCN), only one season is considered as stable reflectance measurements values are provided. Two different periods are considered for LCFR site (leaf-off (October-March) and leaf-on (April-September)) as seasonal dependency is expected in this site. Table 3 summarizes the geospatial statistics for the 4 evaluated RadCalNet sites.

Better agreement between satellite pixels and tower measurements are expected when scores are higher than 2.0 (Cescatti et al., 2012; Sánchez-Zapero et al., 2023, 2020). Based on our analysis of spatial representativeness, better results are expected for GONA. However, it should be noted that the analysis is based on the near infrared and slightly different results could be found for visible channels mainly in sites with some vegetation (LCFR).

Table 3: Geospatial statistics of the four RadCalNet sites for direct validation. Scores are computed for 300 m versus 500 m.

Site	Seasonal Period	Footprint (m)	Tower Height (m)	300m range	500m range	Rcv (%)	Rse (%)	Rst (%)	Rsv (%)	ST_{score}	RAW_{score}
GONA	Single season	126	10	144.2	194.3	-6	33.6	4.7	34.8	2.05	8.40
LCFR	Leaf-off	126	10	60.4	54	-33.4	4.3	-2.5	-10.6	5.04	1.50
	Leaf-on			149.7	347.2	63.1	39.9	-5.4	131.8	0.94	0.79
RVUS	Single season	126	10	46.3	202.9	62.2	6.1	-1.4	338.4	0.71	0.80
BSCN	Single season	19	2	66.2	84	183.8	8.8	-11	26.9	1.21	0.27

2.2.2 HYPERNETS

The HYPERNETS project[2] (Ruddick et al., 2024) aims to ensure that high-quality hyperspectral surface reflectance measurements are available to support the validation of optical Visible and Near InfraRed (VNIR) and Short-wave infrared (SWIR) Copernicus products. Therefore, it established a new autonomous hyperspectral spectroradiometer (HYPSTAR® - www.hypstar.eu) dedicated to land and water surface reflectance validation with instrument-pointing capabilities. In the prototype phase, the instrument is being deployed at 24 sites covering a range of water and land types and a range of climatic and logistic conditions.

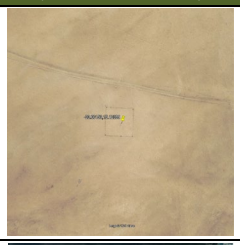
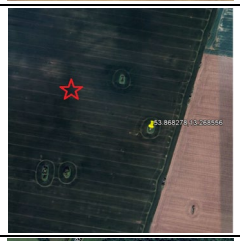
HYPERNETS dataset corresponds to the reflectance L2A products measured with HYPSTAR®-XR sensors every 30 minutes between 9:00 and 17:00 (Universal Time Coordinated - UTC) from different set of geometries (full zenith/azimuth coverage, up to 60° from nadir). The HYPSTAR®-XR (eXtended Range) instruments deployed at each land HYPERNETS site consist of a VNIR and a SWIR sensor and autonomously collect data between 380-1700 nm at various viewing geometries and send it to a central server for quality control and processing. The HYPERNETS processor (Goyens et al., 2021) automatically processes all these data into various products, including the L2A surface reflectance product used here.

The data used in this evaluation corresponds to initial test samples published data (De Vis et al., 2023; Morris et al., 2023; Saberioon et al., 2023) over three selected sites (Table 4) over homogeneous land cover types. Other HYPERNETS sites are not used as they are installed over heterogeneous areas or fields of size less than 100 m x 100 m, which are not suitable for validation of Sentinel-3 products with nominal spatial resolution of 300 m. At the time of this analysis, these datasets are limited in

[2] www.hypernets.eu (access on 14th January 2025)

terms of temporal coverage and angular configuration availability as they are a subset of the complete data record expected as outcome of the project.

Table 4: Characteristics of the three HYPERNETS sites considered in this study.

ID Site	Name (Country)	Latitude; Longitude	Temporal coverage	Land Cover	Reference	Google Earth view (1 km x 1 km)
GHNA	Gobabeb (Namibia)	-23.60153°N; 15.12589° E	May 2022 – April 2023	desert	(De Vis et al., 2023)	
DEGE	Demmin (Germany)	53.868278°N; 13.268556°E(*)	July 2021 – April 2023	agricultural	(Saberioon et al., 2023)	
WWUK	Wytham Woods (UK)	51.777206ºN; 1.338494ºW	April 2022 – April 2023	Deciduous Broadleaf Forest	(Morris et al., 2023)	

(*) For DEGE, the Sentinel-3 extraction coordinate was moved towards the center of the field in order to cover the same field measured by the HYPERNETS (see red star in the Google Earth view).

In summary, HYPERNETS complements the direct validation of Sentinel-3 TOC reflectance V2.3 using the RadCalNet network, which does not cover all surface types and viewing angles (RadCalNet provides only nadir-viewing data). Two assumptions are considered to deal with differences in data acquisition between Sentinel-3 TOC reflectances and HYPERNETS in situ data:

- Similar illumination and viewing configurations are considered. For that, Sentinel-3 and HYPERNETS observations with differences in viewing zenith angles larger than 5º are discarded. Same dates and acquisition times are compared.
- The spectral range of HYPERNETS VNIR sensors is from 380 to 1000 nm (3 nm spectral interval), and the spectral range of HYPERNETS SWIR sensor is from 1000 and 1700 nm (10 nm spectral interval). These narrow-band measurements are made comparable with Sentinel-3 channels using the SRF coefficients of Sentinel-3 [S3-OLCI TG, S3-SLSTR TG].

Note that HYPERNETS measurements do not cover the spectral resolution of SLSTR S6 channel.

2.3 SATELLITE DATASET USED FOR BENCHMARKING: ESA SENTINEL-3 SYN LEVEL-2 (ESA L2 SYN)

The ESA Synergy Level-2 Surface Directional Reflectance (SDR) product [L2-SYN-SDR-processing] is used for benchmarking in direct validation. It provides the atmospherically corrected surface bi-directional reflectance for the sun-reflective SLSTR channels (S1 to S6 for both nadir and oblique views except S4) and for all OLCI channels (except for the oxygen absorption bands Oa13, Oa14, Oa15, and the water vapor bands Oa19 and Oa20) and the corresponding information on aerosol properties.

The ESA L2 SYN SDR products have a resolution of 300 m and are mapped on the same regular grid as OLCI. They are created from a combination of OLCI and SLSTR Level-1B products. The synergy swath is then corresponding to the common area between OLCI and SLSTR nadir view swaths, restricted to the daylight part of the Sentinel-3 orbit (i.e., where the sun zenith angle at satellite ground track is lower than 80°).

The algorithm of L2 SYN processor [S3-L2-SD-03-S02-ATBD] is implemented to estimate surface reflectance from OLCI and SLSTR TOA measurements and is based on four main components: (i) radiative transfer modelling of TOA reflectance to generate look-up tables for a range of aerosol optical depths and aerosol model types, (ii) definition of parameterized models of land surface spectral and angular reflectance, (iii) an inversion algorithm to estimate the optimal atmospheric profile, and (iv) atmospheric correction to estimate surface reflectance for SLSTR and OLCI channels. The core algorithm of this Level-2 SYN processor is an aerosol retrieval based on an approach combining dark object and multiple view-angle methods derived from an existing work for Medium Resolution Imaging Spectrometer (MERIS) and AATSR. Sentinel-3 L2 SYN data is produced since October 2018 for Sentinel-3A and February 2019 for Sentinel-3B. The track record and evolutions of the processing baseline are available at [L2-SYN-processing-baseline].

NOTE: The version used in this study corresponds to the processing baseline version 2 [L2-SYN-processing-baseline] (data access from 1st to 30th February, 2025). The study period where ESA L2 SYN product is used for benchmarking covers the whole 2024 year in the comparison with RadCalNet and the period from July 2021 to April 2023 in the comparison with HYPERNETS.

It should be noted that the recommended ESA calibration factors for OLCI-A and SLSTR are applied in ESA L2 SYN from July 2023 onwards [S3.PN-SYN-L2.10].

The quality assessment of Sentinel-3 L2 SYN SDR products [OMPC.ACR.HBK.003] was performed by indirect validation against the normalized surface reflectance measurements observed from MODIS (MCD43A4.061, a combined Terra and Aqua Product) at 500 m spatial resolution and Visible Infrared Imaging Radiometer Suite (VIIRS) (VNP43MA4.001) at 1 km resolution. Several regional inter-comparisons were performed. A good agreement is observed between both Sentinel 3 A/B derived SYN SDR and MODIS (VIIRS) surface reflectance products, in particular after the correction of discrepancies with respect to acquisition geometry. The observed statistical scores are slightly better with S3B than compared to S3A. There is also a clear influence of surface biome on the intercomparisons results, with better agreement for needle and broad evergreen classes than to cultivated and shrub classes.

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3 RESULTS

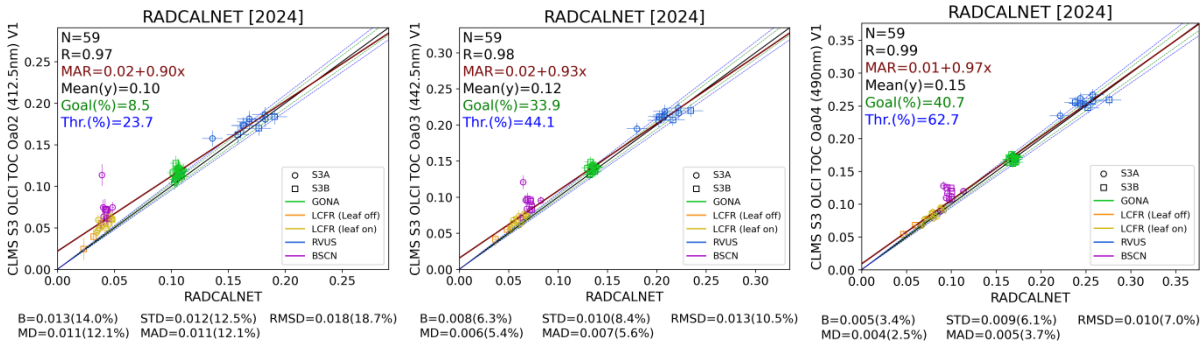
3.1 DIRECT VALIDATION

3.1.1 Error evaluation

3.1.1.1 Comparison with RadCalNet measurements

To estimate the error (APU) of the CLMS Sentinel-3 TOC V1 during the recent 2024 year, scatter-plots and validation metrics versus RadCalNet measurements are produced for OLCI (Figure 2, Table 5) and SLSTR (Figure 4, Table 7) bands. Only Sentinel-3 TOC reflectance retrievals with VZA<10° were used in the comparison, as RadCalNet measurements were acquired at nadir view. Note that only Sentinel-3 pixels showing good quality (see Table 3 of quality assessment report [CGLOPS1_QAR_S3-TOCr300m-v1]) are considered. Error bars in Figure 2 and Figure 4 represent the associated uncertainties of CLMS TOC V1 and RadCalNet measurements. For benchmarking, the equivalent results for ESA L2 SYN products are presented in Figure 3 (Table 6) for OLCI channels and in Figure 5 (Table 8) for SLSTR. Concomitant samples between CLMS and ESA products are used.

Additionally, some relevant statistics (N, relative B and relative RMSD) of the CLMS TOC V1 versus RadCalNet comparison are summarized for different periods in Annex I and Annex II, for OLCI and SLSTR channels respectively. The different periods correspond to the current 2024 year, the cumulative whole period from Jul'18 to Dec'24, the cumulative period from Jul'18 to Jun'22 (i.e., period with geolocation issue) and cumulative period from Jul'22 to Dec'24 (i.e., period without geolocation issue).



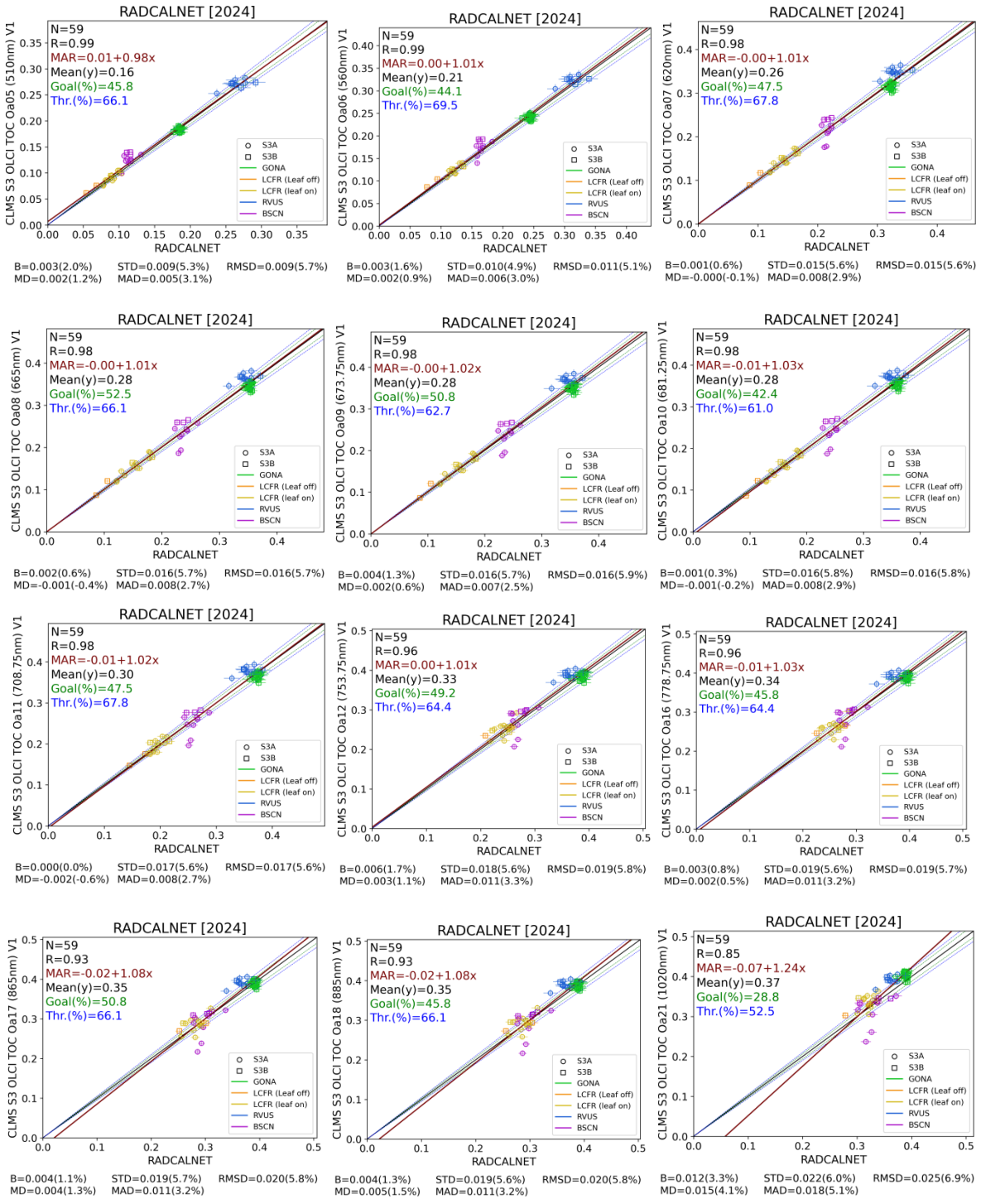
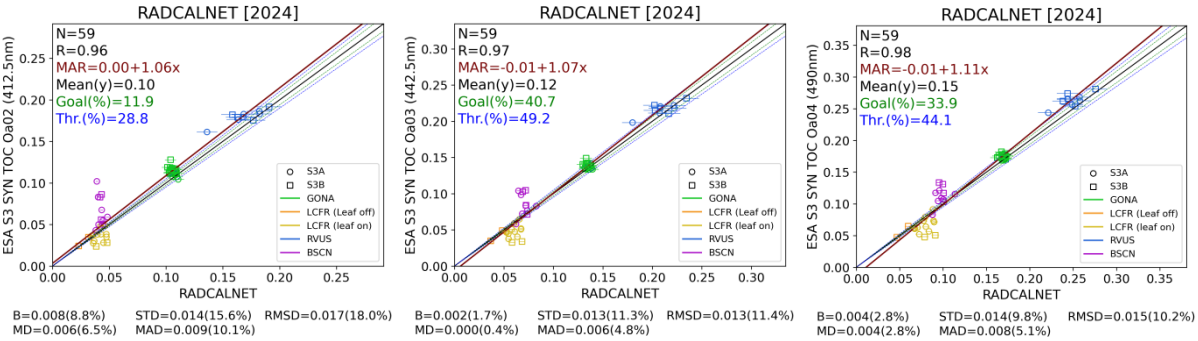


Figure 2: Scatter-plots of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for OLCI channels. Vertical and horizontal bars represent CLMS TOC V1 and RadCalNet associated uncertainties. Green and blue lines delimit goal and threshold requirements.

Table 5: Performance statistics of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for OLCI channels.

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Oa02	59	0.97	0.013 (14.0%)	0.011 (12.1%)	0.012 (12.5%)	0.011 (12.1%)	0.018 (18.7%)	0.02 /0.90	8.5 /23.7
Oa03	59	0.98	0.008 (6.3%)	0.006 (5.4%)	0.010 (8.4%)	0.007 (5.6%)	0.013 (10.5%)	0.02 /0.93	33.9 /44.1
Oa04	59	0.99	0.005 (3.4%)	0.004 (2.5%)	0.009 (6.1%)	0.005 (3.7%)	0.010 (7.0%)	0.01 /0.97	40.7 /62.7
Oa05	59	0.99	0.003 (2.0%)	0.002 (1.2%)	0.009 (5.3%)	0.005 (3.1%)	0.009 (5.7%)	0.01 /0.98	45.8 /66.1
Oa06	59	0.99	0.003 (1.6%)	0.002 (0.9%)	0.010 (4.9%)	0.006 (3.0%)	0.011 (5.1%)	0.00 /1.01	44.1 /69.5
Oa07	59	0.98	0.001 (0.6%)	>-0.001 (-0.1%)	0.015 (5.6%)	0.008 (2.9%)	0.015 (5.6%)	0.00 /0.98	47.5 /67.8
Oa08	59	0.98	0.002 (0.6%)	-0.001 (-0.4%)	0.016 (5.7%)	0.008 (2.7%)	0.016 (5.7%)	0.00 /1.01	52.5 /66.1
Oa09	59	0.98	0.004 (1.3%)	0.002 (0.6%)	0.016 (5.7%)	0.007 (2.5%)	0.016 (5.9%)	0.00 /1.02	50.8 /62.7
Oa10	59	0.98	0.001 (0.3%)	-0.001 (-0.2%)	0.016 (5.8%)	0.008 (2.9%)	0.016 (5.8%)	-0.01 /1.03	42.4 /61.0
Oa11	59	0.98	0.000 (0.0%)	-0.002 (-0.6%)	0.017 (5.6%)	0.008 (2.7%)	0.017 (5.6%)	-0.01 /1.02	47.5 /67.8
Oa12	59	0.96	0.006 (1.7%)	0.003 (1.1%)	0.018 (5.6%)	0.011 (3.3%)	0.019 (5.8%)	0.00 /1.01	49.2 /64.4
Oa16	59	0.96	0.003 (0.8%)	0.002 (0.5%)	0.019 (5.6%)	0.011 (3.2%)	0.019 (5.7%)	-0.01 /1.03	45.8 /64.4
Oa17	59	0.93	0.004 (1.1%)	0.004 (1.3%)	0.019 (5.7%)	0.011 (3.2%)	0.020 (5.8%)	-0.02 /1.08	50.8 /66.1
Oa18	59	0.93	0.004 (1.3%)	0.003 (1.5%)	0.019 (5.6%)	0.011 (3.2%)	0.020 (5.8%)	-0.02 /1.08	45.8 /66.1
Oa21	59	0.85	0.012 (3.3%)	0.015 (4.1%)	0.022 (6.0%)	0.018 (5.1%)	0.025 (6.9%)	-0.07 /1.24	28.8 /52.5



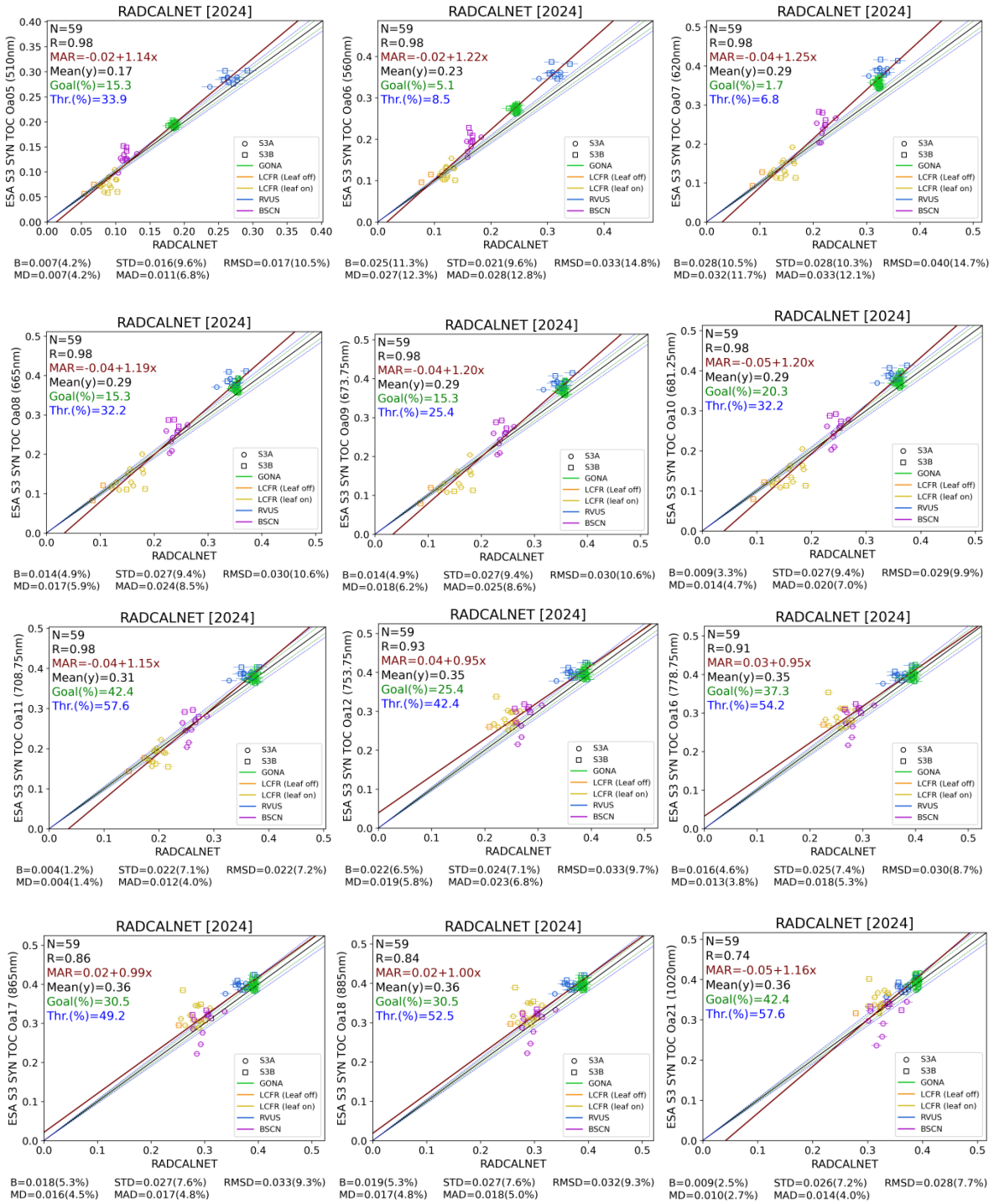


Figure 3: Scatter-plots of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for OLCI channels. Green and blue lines delimit goal and threshold requirements.

Table 6: Performance statistics of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for OLCI channels.

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Oa02	59	0.96	0.008 (8.8%)	0.006 (6.5%)	0.014 (15.6%)	0.009 (10.1%)	0.017 (18.0%)	0.00 /1.06	11.9 /28.8
Oa03	59	0.97	0.002 (1.7%)	0.000 (11.3%)	0.013 (11.3%)	0.006 (4.8%)	0.013 (11.4%)	-0.01 /1.07	40.7 /49.2
Oa04	59	0.98	0.004 (2.8%)	0.004 (2.8%)	0.014 (9.8%)	0.008 (5.1%)	0.015 (10.2%)	-0.01 /1.11	33.9 /44.1
Oa05	59	0.98	0.007 (4.2%)	0.007 (4.2%)	0.016 (9.6%)	0.011 (6.8%)	0.017 (10.5%)	-0.02 /1.14	15.3 /33.9
Oa06	59	0.98	0.025 (11.3%)	0.027 (12.3%)	0.021 (9.6%)	0.028 (12.8%)	0.033 (14.8%)	-0.02 /1.22	5.1 /8.5
Oa07	59	0.98	0.028 (10.5%)	0.032 (11.7%)	0.028 (10.3%)	0.033 (12.1%)	0.040 (14.7%)	-0.04 /1.25	1.7 /6.8
Oa08	59	0.98	0.014 (4.9%)	0.017 (5.9%)	0.027 (9.4%)	0.024 (8.5%)	0.030 (10.6%)	-0.04 /1.19	15.3 /32.2
Oa09	59	0.98	0.014 (4.9%)	0.018 (6.2%)	0.027 (9.4%)	0.025 (8.6%)	0.030 (10.6%)	-0.04 /1.20	15.3 /25.4
Oa10	59	0.98	0.009 (3.3%)	0.014 (4.7%)	0.027 (9.4%)	0.020 (7.0%)	0.029 (9.9%)	-0.05 /1.20	20.3 /32.2
Oa11	59	0.98	0.004 (1.2%)	0.004 (1.4%)	0.022 (7.1%)	0.012 (4.0%)	0.022 (7.2%)	-0.04 /1.15	42.4 /57.6
Oa12	59	0.93	0.022 (6.5%)	0.019 (5.8%)	0.024 (7.1%)	0.023 (6.8%)	0.033 (9.7%)	0.04 /0.95	25.4 /42.4
Oa16	59	0.16	0.016 (4.6%)	0.013 (3.8%)	0.025 (7.4%)	0.018 (5.3%)	0.030 (8.7%)	0.03 /0.95	37.3 /54.2
Oa17	59	0.86	0.018 (5.3%)	0.016 (4.5%)	0.027 (7.6%)	0.017 (4.8%)	0.033 (9.3%)	0.02 /0.99	30.5 /49.2
Oa18	59	0.84	0.019 (5.3%)	0.017 (4.8%)	0.027 (7.6%)	0.018 (5.0%)	0.032 (9.3%)	0.02 /1.00	30.5 /52.5
Oa21	59	0.74	0.009 (2.5%)	0.010 (2.7%)	0.026 (7.2%)	0.014 (4.0%)	0.028 (7.7%)	-0.05 /1.16	42.4 /57.6

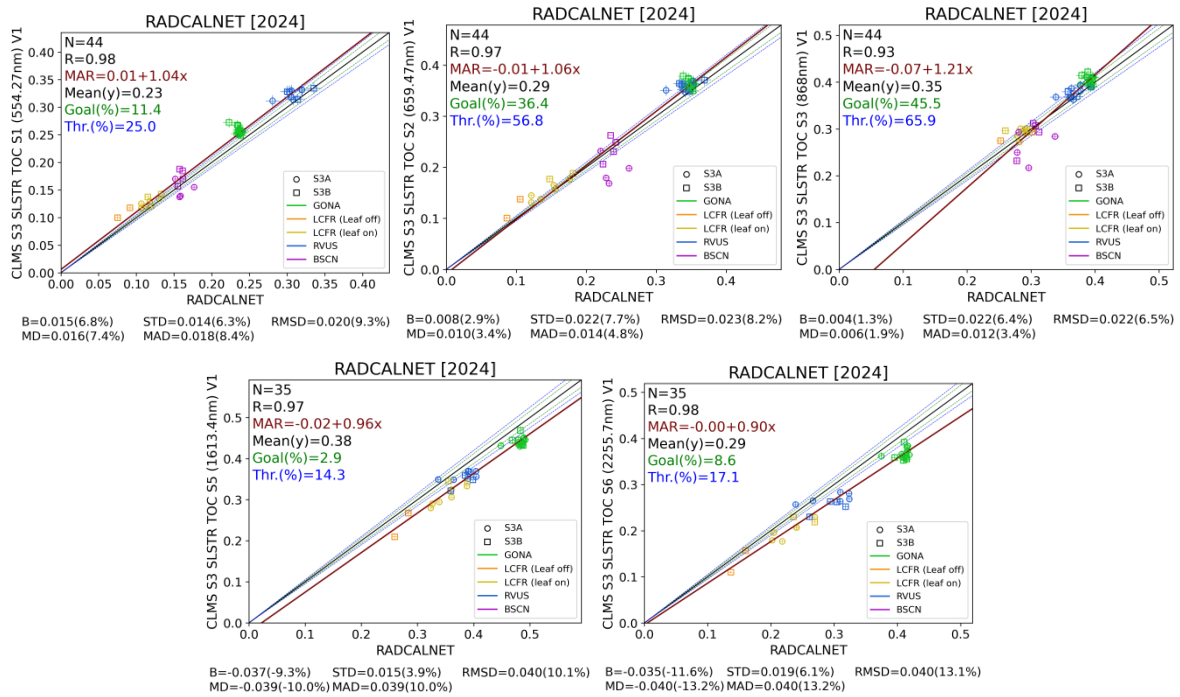


Figure 4: Scatter-plots of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for SLSTR channels. Vertical and horizontal bars represent CLMS TOC V1 and RadCalNet associated uncertainties. Green and blue lines delimit goal and threshold requirements.

Table 7: Performance statistics of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for SLSTR channels. BSCN does not provide measurements for spectral ranges of S5 and S6.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
S1	44	0.98	0.015 (6.8%)	0.016 (7.4%)	0.014 (6.3%)	0.018 (8.4%)	0.020 (9.3%)	0.01 /1.04	11.4 /25.0
S2	44	0.97	0.008 (2.9%)	0.010 (3.4%)	0.022 (7.7%)	0.014 (4.8%)	0.023 (8.2%)	-0.01 /1.06	36.4 /56.8
S3	44	0.93	0.004 (1.3%)	0.006 (1.9%)	0.022 (6.4%)	0.012 (3.4%)	0.022 (6.5%)	-0.07 /1.21	45.5 /65.9
S5	35	0.97	-0.037 (-9.3%)	-0.039 (-10.0%)	0.015 (3.9%)	0.039 (10.0%)	0.040 (10.1%)	-0.02 /0.96	2.9 /14.3
S6	35	0.97	-0.035 (-11.6%)	-0.040 (-13.2%)	0.019 (6.1%)	0.040 (13.2%)	0.040 (13.1%)	0.00 /0.90	8.6 /17.1

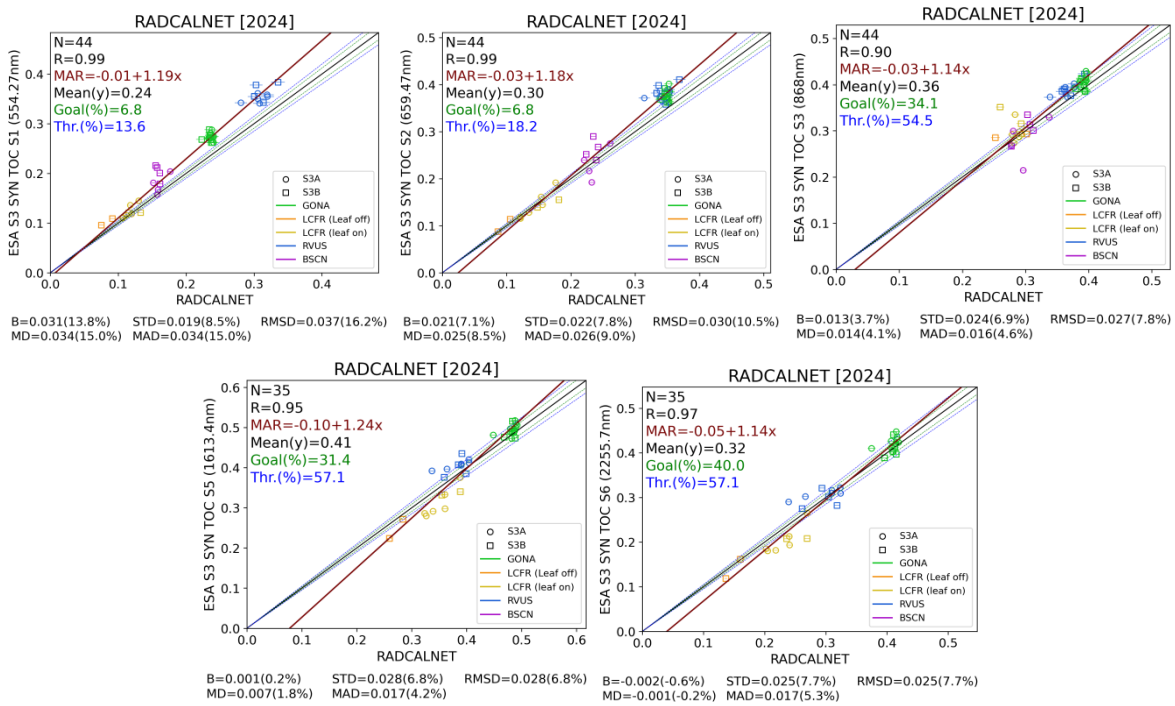


Figure 5: Scatter-plots of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for SLSTR channels. Green and blue lines delimit goal and threshold requirements.

Table 8: Performance statistics of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for SLSTR channels. BSCN does not provide measurements for spectral ranges of S5 and S6.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
S1	44	0.99	0.031 (13.8%)	0.034 (15.0%)	0.019 (8.5%)	0.034 (15.0%)	0.037 (16.2%)	-0.01 /1.19	6.8 /13.6
S2	44	0.99	0.021 (7.1%)	0.025 (8.5%)	0.022 (7.8%)	0.026 (9.0%)	0.030 (10.5%)	-0.03 /1.18	6.8 /18.2
S3	44	0.90	0.013 (3.7%)	0.014 (4.1%)	0.024 (6.9%)	0.016 (4.6%)	0.027 (7.8%)	-0.03 /1.14	34.1 /54.5
S5	35	0.95	0.001 (0.2%)	0.007 (1.8%)	0.028 (6.8%)	0.017 (4.2%)	0.028 (6.8%)	-0.10 /1.24	31.4 /57.1
S6	35	0.97	-0.002 (-0.6%)	-0.001 (-0.2%)	0.025 (7.7%)	0.017 (5.3%)	0.025 (7.7%)	-0.05 /1.14	40.0 /57.1

Main conclusions for Sentinel-3 OLCI channels in 2024 are:

- CLMS TOC V1 (Figure 2, Table 5) shows:

    - Remarkably good accuracy with almost no bias or very low bias (i.e., B and MD typically within ±2%) for most channels [Oa05 – Oa18] and larger positive bias for the lower spectral channels (B of 12.5%, 8.4% and 2.5% for Oa02, Oa03 and Oa04) and higher (3.3% for Oa21).
    - Overall uncertainty (RMSD) typically better than 6% is found for most channels [Oa05 – Oa18].
- Regarding the comparison of CLMS TOC V1 over different periods (Annex I):
    - For the accuracy, it seems that the number of samples has a significant impact on the statistics, especially for the low spectral bands. Consequently, no general trend is obtained.
    - At the opposite, in terms of uncertainty, 2024 shows significantly better results (RMSD around 5-6% for most channels) than cumulative periods (RMSD around 9-15% for most channels) while the period July 2018-June 2022, fully affected by the geolocation issue, display the worse results.
- CLMS TOC V1 shows significantly better results than ESA L2 SYN (Figure 3, Table 6): systematic positive biases for ESA product typically between 3% and 12% and RMSD typically from 7.2% to 14.8% (Oa06) (larger for Oa02, RMSD=18%). The systematic positive bias of ESA L2 SYN for all OLCI channels is in line (but higher in magnitude) to that reported for ESA level 1 radiometric annual performance report for 2023 year [OMPC.ACR.APR.004], where OLCI radiances showed an excess of brightness (up to 5%).

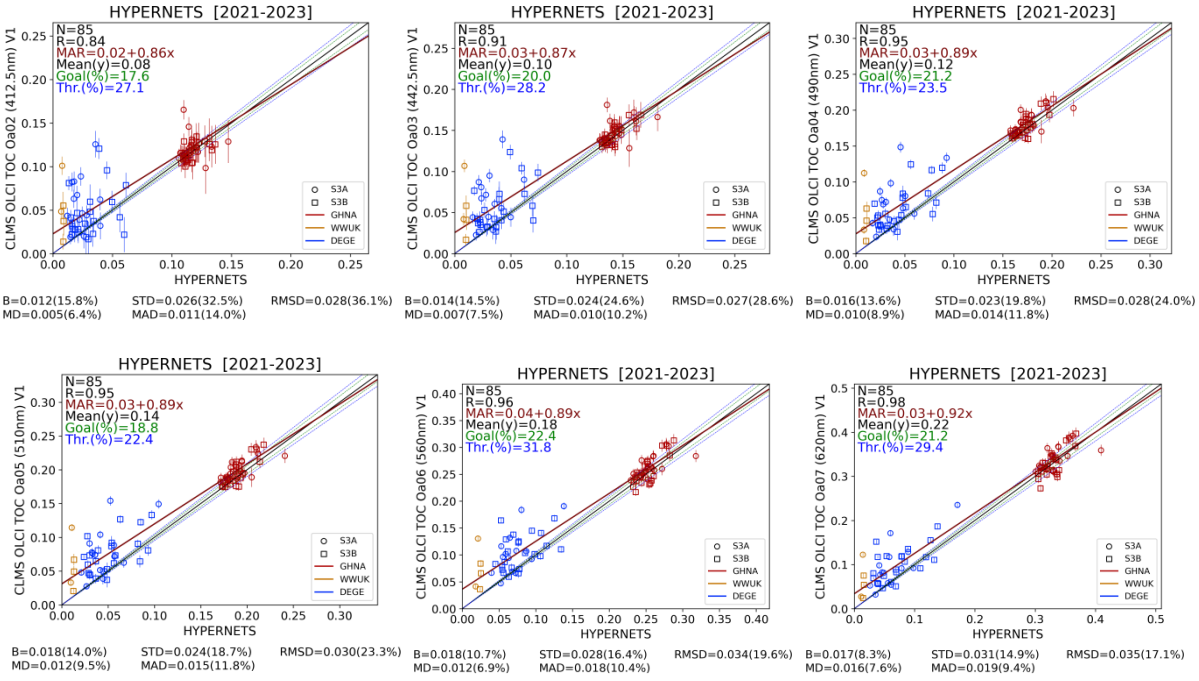
Main conclusions for Sentinel-3 SLSTR channels are:

- CLMS TOC V1 (Figure 4, Table 7) shows:
    - Accuracy (B, MD) of around 7% and 3% for S1 and S2 visible channels and lower than 2% for S3 NIR channel, showing positive mean differences.
    - Worse accuracy (B, MD) for SLSTR channels in SWIR domain, with negative mean differences of around -9% and -12% for S5 and S6, as expected since ESA recommended SLSTR calibration coefficients are not applied in CLMS TOC V1 processing chain.
    - Overall uncertainty (RMSD) ranges from 6.5% (S3) to 13.1% (S6).
- The direct validation of CLMS TOC V1 for different periods (Annex II) shows:
    - In terms of accuracy, slightly worse results are found in 2024, compared with cumulative periods, for VNIR channels and slightly better results for SWIR S5 and S6 channels.
    - In terms of overall uncertainty, better results are found in 2024 (RMSD ~ 6%-13%), compared to other periods (RMSD ~ 9%-16%).
- CLMS TOC V1 shows better performance than ESA L2 SYN products (Figure 5, Table 8) VNIR channels (B of 13.8%, 7.1% and 3.7% for S1, S2 and S3). ESA L2 SYN shows,

however better agreement for SWIR S5 and S6 channels (B of 6.8% and 7.7%) as expected as ESA calibration adjustment factors for SLSTR were incorporated in ESA L2 SYN products from July 2023 onwards [S3.PN-SYN-L2.10].

3.1.1.2 Comparison with HYPERNETS measurements

Scatter-plots and validation metrics of Sentinel-3 CLMS TOC V1 versus HYPERNETS measurements are performed for OLCI (Figure 6, Table 9) and SLSTR (Figure 8, Table 11) channels for the period of 2021-2023 (limited to HYPERNETS availability as no data is available for 2024 year, see Table 4). Sentinel-3 TOC reflectance retrievals and HYPERNETS observations are compared under similar viewing geometries, only for observations with differences in viewing angles lower than 5°. Note that only Sentinel-3 pixels showing good quality (see Table 3 of quality assessment report [CGLOPS1_QAR_S3-TOCr300m-v1]) are considered. For benchmarking, the equivalent results for ESA L2 SYN products are presented in Figure 7 (Table 10) for OLCI channels and in Figure 9 (Table 12) for SLSTR. Concomitant samples between CLMS and ESA products are used.



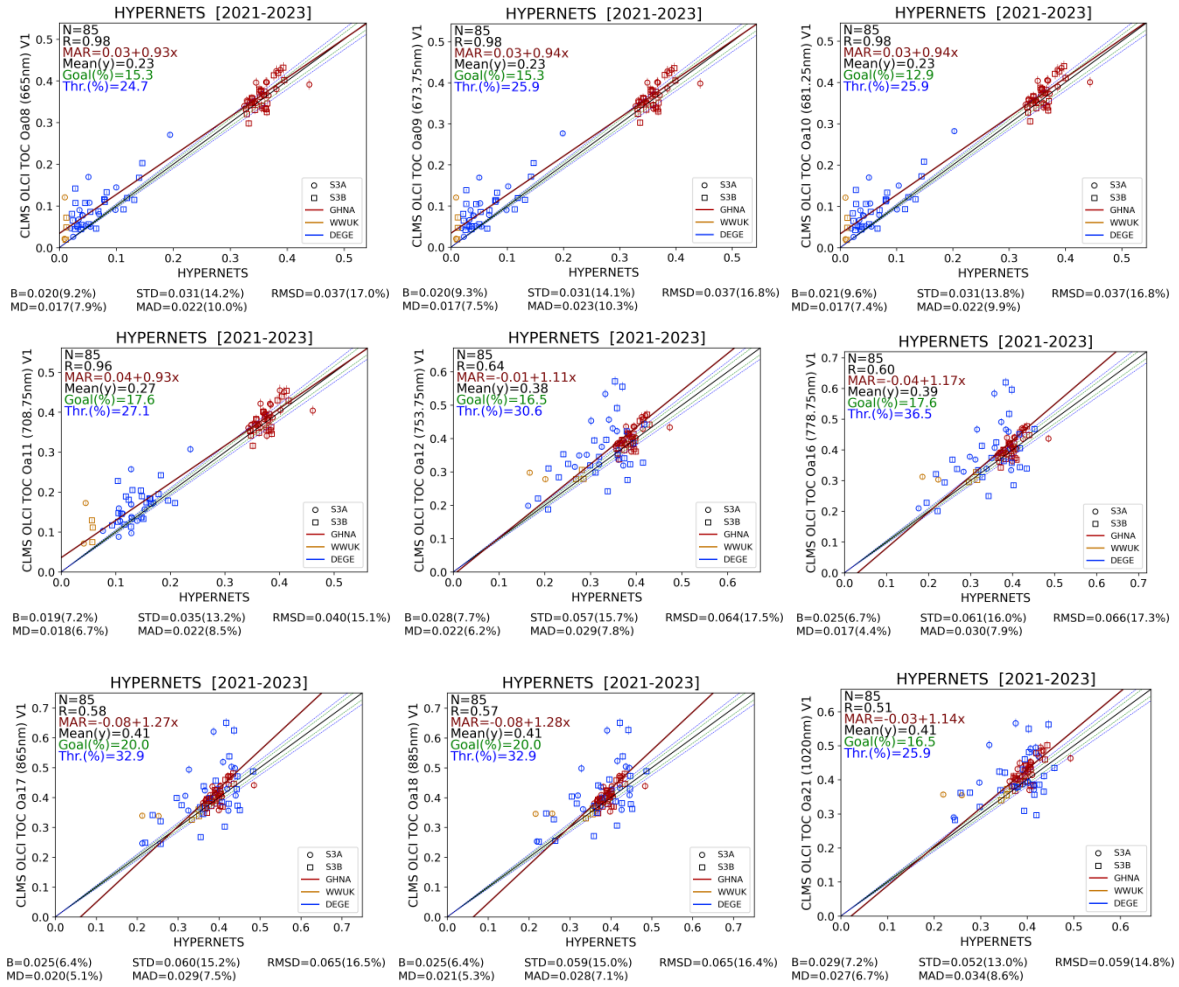
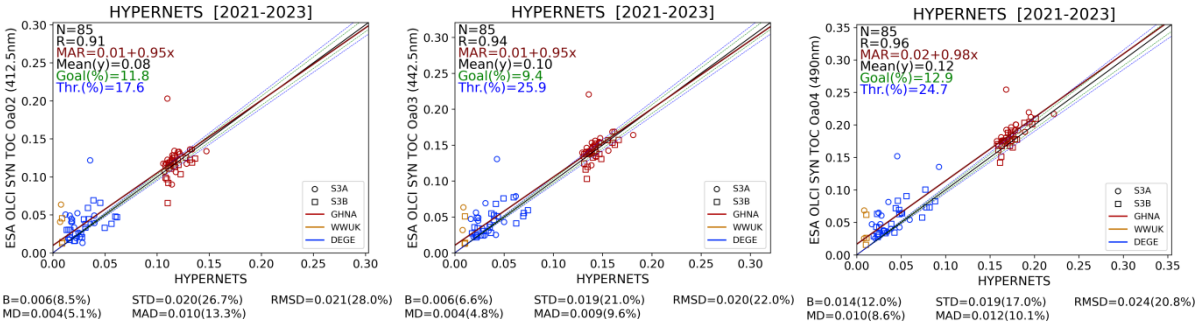


Figure 6: Scatter-plots of CLMS Sentinel-3 TOC V1 versus HYPERNETS measurements (ΔVZA<5º; 2021-2023) for OLCI channels. Vertical bars represent CLMS TOC V1 associated uncertainties. Green and blue lines delimit goal and threshold requirements.

Table 9: Performance statistics of CLMS Sentinel-3 TOC V1 versus HYPERNETS measurements (ΔVZA<5°; 2021-2023) for OLCI channels.

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Oa02	85	0.84	0.012 (15.8%)	0.005 (6.4%)	0.026 (32.5%)	0.011 (14.0%)	0.028 (36.1%)	0.02 /0.86	17.6 /27.1
Oa03	85	0.91	0.014 (14.5%)	0.007 (7.5%)	0.024 (24.6%)	0.010 (10.2%)	0.027 (28.6%)	0.03 /0.87	20.0 /28.2
Oa04	85	0.95	0.016 (13.6%)	0.010 (8.9%)	0.023 (19.8%)	0.014 (11.8%)	0.028 (24.0%)	0.03 /0.89	21.2 /23.5
Oa05	85	0.95	0.018 (14.0%)	0.012 (9.5%)	0.024 (18.7%)	0.015 (11.8%)	0.030 (23.3%)	0.03 /0.89	18.8 /22.4
Oa06	85	0.96	0.018 (10.7%)	0.012 (6.9%)	0.028 (16.4%)	0.018 (10.4%)	0.034 (19.6%)	0.04 /0.89	22.4 /31.8
Oa07	85	0.98	0.017 (8.3%)	0.016 (7.6%)	0.031 (14.9%)	0.019 (9.4%)	0.035 (17.1%)	0.03 /0.92	21.1 /29.4
Oa08	85	0.98	0.020 (9.2%)	0.017 (7.9%)	0.031 (14.2%)	0.022 (10.0%)	0.037 (17.0%)	0.03 /0.93	15.3 /24.7
Oa09	85	0.98	0.020 (9.3%)	0.017 (7.5%)	0.031 (14.1%)	0.023 (10.3%)	0.037 (16.8%)	0.03 /0.94	15.3 /25.9
Oa10	85	0.98	0.021 (9.6%)	0.017 (7.4%)	0.031 (13.8%)	0.022 (9.9%)	0.037 (16.8%)	0.03 /0.94	12.9 /25.9
Oa11	85	0.96	0.019 (7.2%)	0.018 (6.7%)	0.035 (13.2%)	0.022 (8.5%)	0.040 (15.1%)	0.04 /0.93	17.6 /27.1
Oa12	85	0.64	0.028 (7.7%)	0.022 (6.2%)	0.057 (15.7%)	0.029 (7.8%)	0.064 (17.5%)	-0.01 /1.11	16.5 /30.6
Oa16	85	0.60	0.025 (6.7%)	0.017 (4.4%)	0.061 (16.0%)	0.030 (7.9%)	0.066 (17.3%)	-0.04 /1.17	17.6 /36.5
Oa17	85	0.58	0.025 (6.4%)	0.020 (5.1%)	0.060 (15.2%)	0.029 (7.5%)	0.065 (16.5%)	-0.08 /1.27	20.0 /32.9
Oa18	85	0.57	0.025 (6.4%)	0.021 (5.3%)	0.059 (15.0%)	0.028 (7.1%)	0.065 (16.4%)	-0.08 /1.28	20.0 /32.9
Oa21	85	0.51	0.029 (7.2%)	0.027 (6.7%)	0.052 (13.0%)	0.034 (8.6%)	0.059 (14.8%)	-0.03 /1.14	16.5 /25.9



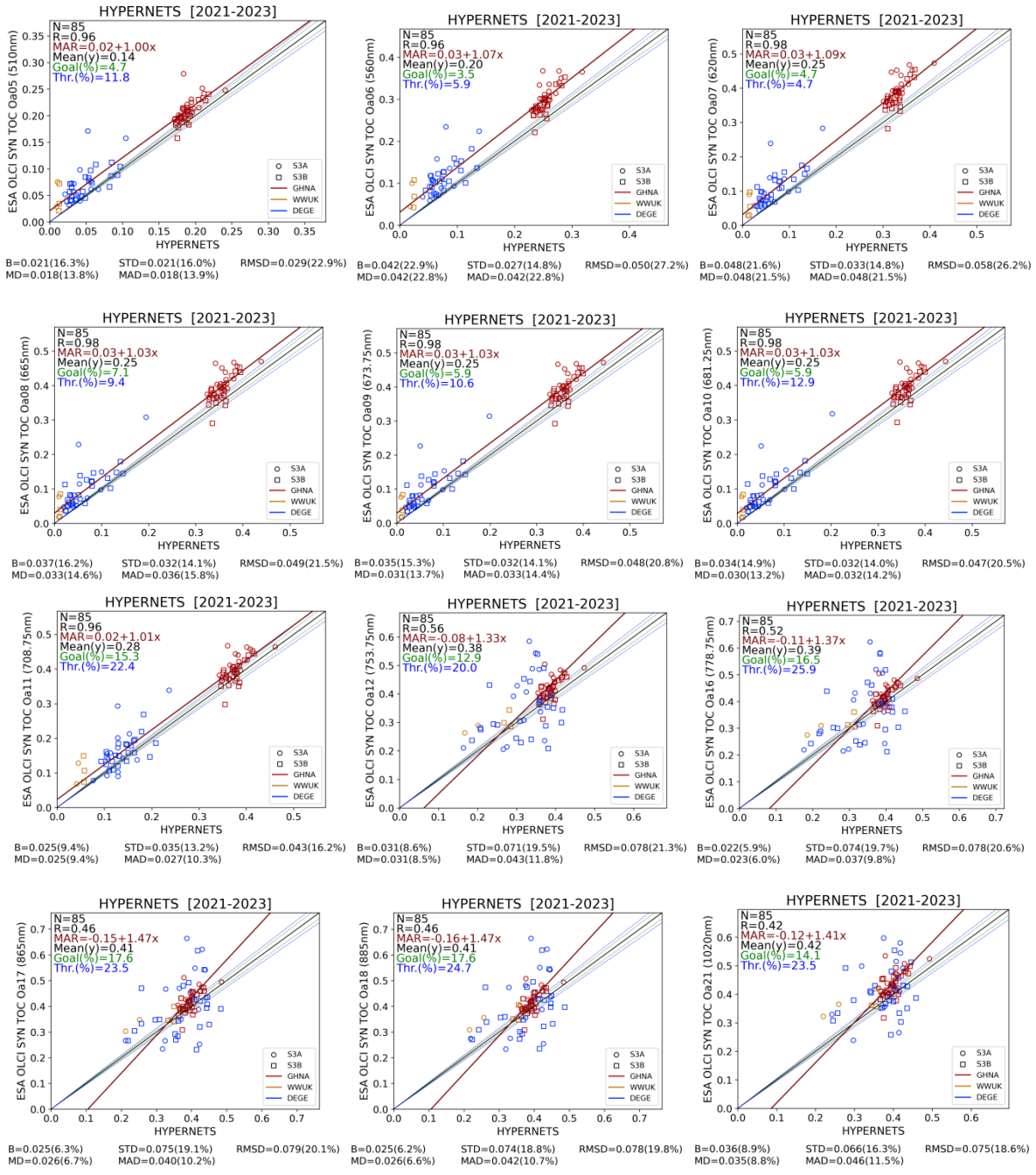


Figure 7: Scatter-plots of ESA Sentinel-3 L2 SYN versus HYPERNETS measurements (ΔVZA<5º; 2021-2023) for OLCI channels. Green and blue lines delimit goal and threshold requirements.

Table 10: Performance statistics of ESA Sentinel-3 L2 SYN versus HYPERNETS measurements (ΔVZA<5°; 2021-2023) for OLCI channels.

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Oa02	85	0.91	0.006 (8.5%)	0.004 (5.1%)	0.020 (26.7%)	0.010 (13.3%)	0.021 (28.0%)	0.01 /0.95	11.8 /17.6
Oa03	85	0.94	0.006 (6.6%)	0.004 (4.8%)	0.019 (21.0%)	0.009 (9.6%)	0.020 (22.0%)	0.01 /0.95	9.4 /25.9
Oa04	85	0.96	0.014 (12.0%)	0.010 (8.6%)	0.019 (17.0%)	0.012 (10.1%)	0.024 (20.8%)	0.02 /0.98	12.9 /24.7
Oa05	85	0.96	0.021 (16.3%)	0.018 (13.8%)	0.021 (16.0%)	0.018 (13.9%)	0.029 (22.9%)	0.02 /1.00	4.7 /11.8
Oa06	85	0.96	0.042 (22.9%)	0.042 (22.8%)	0.027 (14.8%)	0.042 (22.8%)	0.050 (27.2%)	0.03 /1.07	3.5 /5.9
Oa07	85	0.98	0.048 (21.6%)	0.048 (21.5%)	0.033 (14.8%)	0.048 (21.5%)	0.058 (26.2%)	0.03 /1.09	4.7 /4.7
Oa08	85	0.98	0.037 (16.2%)	0.033 (14.6%)	0.032 (14.1%)	0.036 (15.8%)	0.049 (21.5%)	0.03 /1.03	7.1 /9.4
Oa09	85	0.98	0.035 (15.3%)	0.031 (13.7%)	0.032 (14.1%)	0.033 (14.4%)	0.048 (20.8%)	0.03 /1.03	5.9 /10.6
Oa10	85	0.98	0.034 (14.9%)	0.030 (13.2%)	0.032 (14.0%)	0.032 (14.2%)	0.047 (20.5%)	0.03 /1.03	5.9 /12.9
Oa11	85	0.96	0.025 (9.4%)	0.025 (9.4%)	0.035 (13.2%)	0.027 (10.3%)	0.043 (16.2%)	0.02 /1.01	15.3 /2.4
Oa12	85	0.56	0.031 (8.6%)	0.031 (8.5%)	0.071 (19.5%)	0.043 (11.8%)	0.078 (21.3%)	-0.08 /1.33	12.9 /20.0
Oa16	85	0.52	0.022 (5.9%)	0.023 (6.0%)	0.074 (19.7%)	0.037 (9.8%)	0.078 (20.6%)	-0.11 /1.37	16.5 /25.9
Oa17	85	0.46	0.025 (6.3%)	0.026 (6.7%)	0.075 (19.1%)	0.040 (10.2%)	0.079 (20.1%)	-0.15 /1.47	17.6 /26.5
Oa18	85	0.46	0.025 (6.2%)	0.026 (6.6%)	0.074 (18.8%)	0.042 (10.7%)	0.078 (19.8%)	-0.15 /1.47	17.6 /24.7
Oa21	85	0.42	0.036 (8.9%)	0.035 (8.8%)	0.066 (16.3%)	0.046 (11.5%)	0.075 (18.6%)	-0.12 /1.41	14.1 /23.5

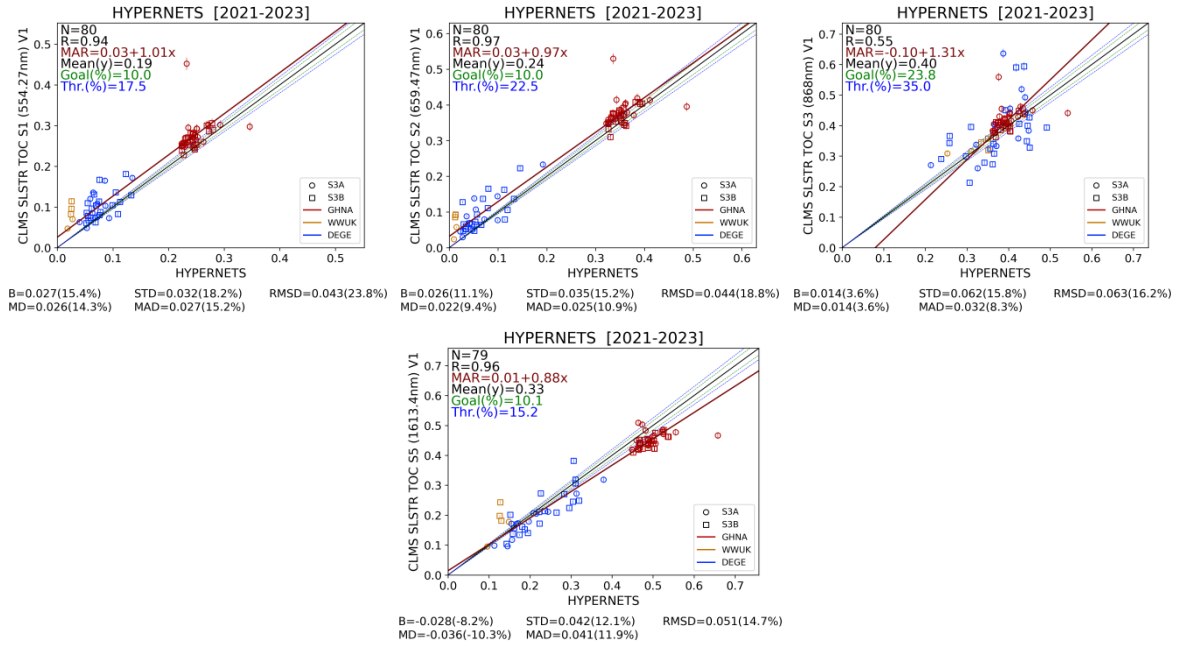


Figure 8: Scatter-plots of CLMS Sentinel-3 TOC V1 versus HYPERNETS measurements (ΔVZA<5º; 2021-2023) for SLSTR channels. Vertical bars represent CLMS TOC V1 associated uncertainties. Green and blue lines delimit goal and threshold requirements.

Table 11: Performance statistics of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (2021-2023) for SLSTR channels.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
S1	80	0.94	0.027 (15.4%)	0.026 (14.3%)	0.032 (18.2%)	0.027 (15.2%)	0.043 (23.8%)	0.03 /1.01	10.0 /17.5
S2	80	0.97	0.026 (11.1%)	0.022 (9.4%)	0.035 (15.2%)	0.025 (10.9%)	0.044 (18.8%)	0.03 /0.97	10.0 /22.5
S3	80	0.55	0.014 (3.6%)	0.014 (3.6%)	0.062 (15.8%)	0.032 (8.3%)	0.063 (16.2%)	-0.10 /1.31	23.8 /35.0
S5	79	0.96	-0.028 (-8.2%)	-0.036 (-10.3%)	0.042 (12.1%)	0.041 (11.9%)	0.051 (14.7%)	0.01 /0.88	10.1 /15.2

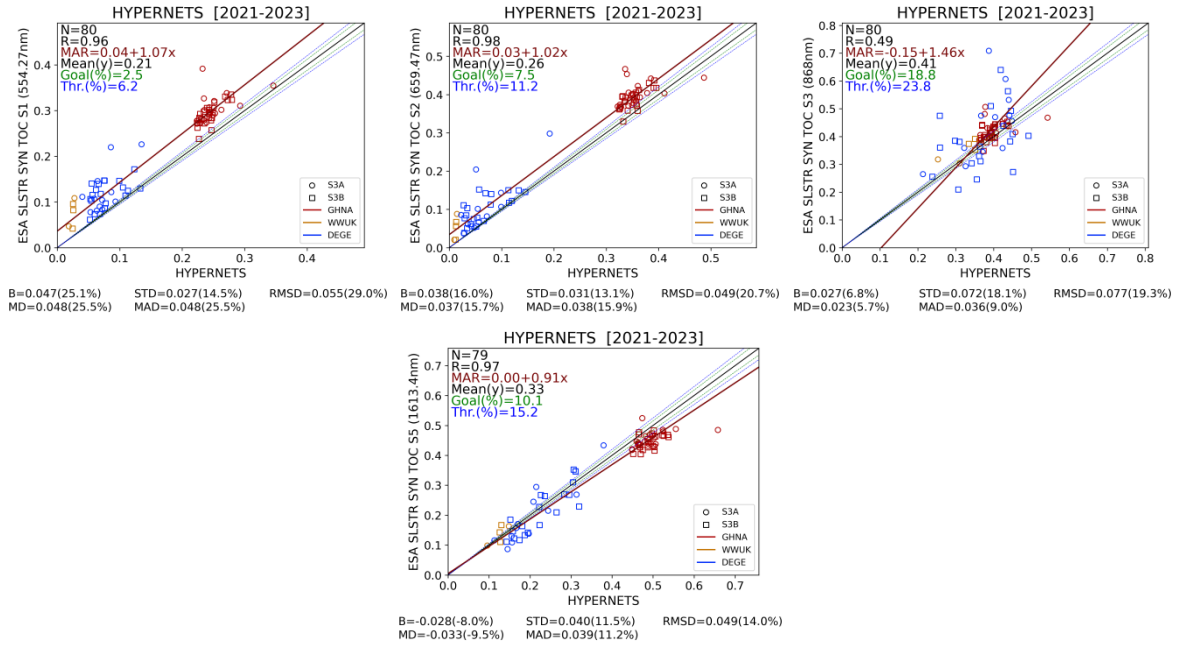


Figure 9: Scatter-plots of ESA Sentinel-3 L2 SYN versus HYPERNETS measurements (ΔVZA<5º; 2021-2023) for SLSTR channels. Green and blue lines delimit goal and threshold requirements.

Table 12: Performance statistics of ESA Sentinel-3 L2 SYN versus HYPERNETS measurements (ΔVZA<5º; 2021-2023) for SLSTR channels.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
S1	80	0.96	0.047 (25.1%)	0.048 (25.5%)	0.027 (14.5%)	0.048 (25.5%)	0.055 (29.0%)	0.04 /1.07	2.5 /6.2
S2	80	0.98	0.038 (16.0%)	0.037 (15.7%)	0.031 (13.1%)	0.038 (15.9%)	0.049 (20.7%)	0.03 /1.02	7.5 /11.2
S3	80	0.49	0.027 (6.8%)	0.023 (5.7%)	0.072 (18.1%)	0.036 (9.0%)	0.077 (19.3%)	-0.15 /1.46	18.8 /23.8
S5	79	0.97	-0.028 (-8.0%)	-0.033 (-9.5%)	0.040 (11.5%)	0.039 (11.2%)	0.049 (14.0%)	0.00 /0.91	10.1 /15.2

Main conclusions for the comparison of Sentinel-3 products versus HYPERNETS for OLCI channels are:

- CLMS TOC V1 (Figure 6, Table 9) shows:
  - Positive median deviation (MD) ranging from 4.4% (Oa16) up to 9.5% (Oa05). Note that MD is more suitable than B in those cases due to the influence of several outliers over this limited sampling.

- Overall uncertainty (RMSD) around 15%-20% for most channels [Oa06-Oa21], showing large values (from 23% to 36%) for the lower spectral bands.

- CLMS TOC V1 shows typically better results than ESA L2 SYN (Figure 7, Table 10) for most channels: accuracy (MD) of ESA product ranging from 4.8% (Oa03) up to 22.8% (Oa06%) and RMSD values around 16%-28%.

Main conclusions for Sentinel-3 SLSTR products versus HYPERNETS measurements are:

- CLMS TOC V1 (Figure 7, Table 10) shows large positive deviation for visible domain (MD of 14.3% and 9.4% for S1 and S2) and positive deviation for NIR (MD of 3.6% for S3). Large negative deviation (MD=-10.3%) is found for S5 channel in SWIR, as expected due to the lack of ESA correction factors of CLMS TOC V1 processing chain.
- ESA L2 SYN (Figure 9, Table 12) shows worse agreement for VNIR S1, S2 and S3 channels, with larger positive deviation of 25.5%, 15.7% and 5.7%. The performance against HYPERNETS for S5 channel is similar to that found for CLMS TOC V1. Note that for the availability period of HYPERNETS measurements (July 2021 – April 2023), ESA correction calibration factors are not applied in ESA L2 SYN product.

3.1.2 Conformity testing

3.1.2.1 Simple acceptance

The summary of the percentages of cases within goal and threshold uncertainty requirements in the comparison of Sentinel-3 CLMS TOC V1 and ESA L2 SYN satellite products versus RadCalNet and HYPERNETS measurements are summarized in Figure 10 and Figure 11.

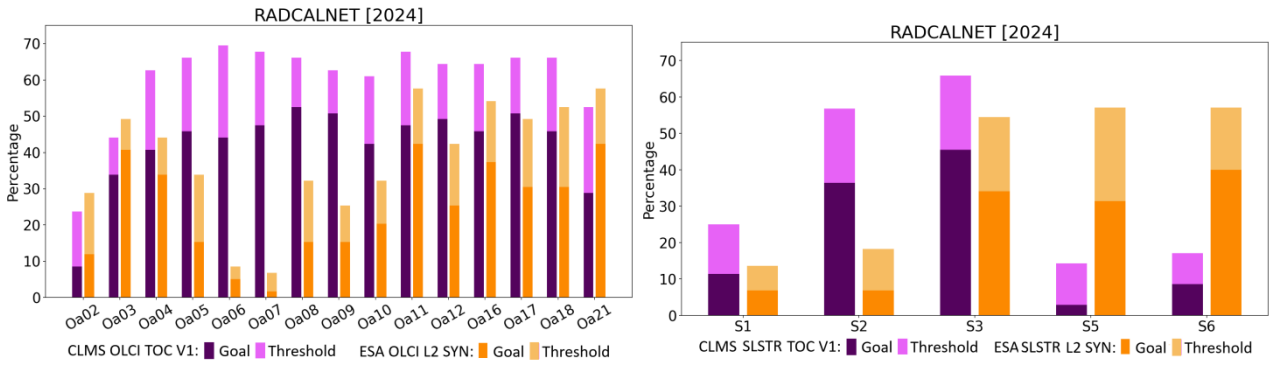


Figure 10: Percentage of cases within goal (dark colour) and threshold (light colour) requirements in the comparison of CLMS TOC V1 (purple) and ESA L2 SYN (orange) satellite products (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for OLCI (left) and SLSTR (right) channels.

Main conclusions from the simple acceptance conformity testing of the comparison of Sentinel-3 based products versus RadCalNet measurements in 2024 (Figure 10):

- CLMS TOC V1 shows typically around 40-50% of pixels within goal requirement (i.e., 3%) for most OLCI channels and typically 55-70% considering threshold level (5%), except for Oa02 and Oa03 with lower percentages. Significantly better results are found compared with ESA L2 SYN.
- In case of SLSTR, satisfactory results of CLMS TOC V1 are found for S2 and S3 channels, with around 40-45% of pixels within goal and 60-65% within threshold. On the contrary, lower compliance is found for S1, S5 and S6 (<12% goal and <25% threshold). CLMS V1 shows better results than ESA L2 SYN for VNIR channels (S1, S2 and S3) and, as expected, and worse results for S5 and S6 (>55% within threshold) due to the application of ESA correction calibration factors in the ESA L2 SYN processing.

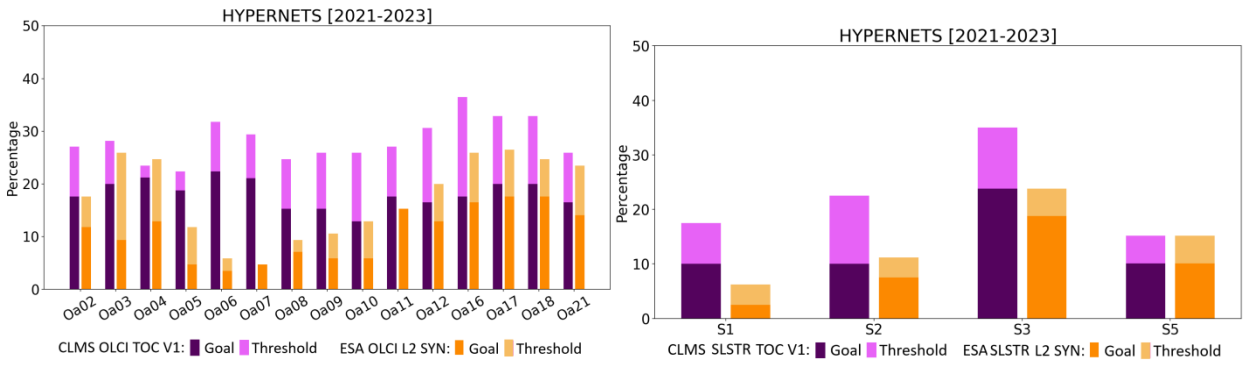


Figure 11: Percentage of cases within goal (dark colour) and threshold (light colour) requirements in the comparison of CLMS TOC V1 (purple) and ESA L2 SYN (orange) satellite products versus HYPERNETS measurements (ΔVZA<5º; 2021-2023) for OLCI (left) and SLSTR (right) channels.

The conformity testing with HYPERNETS measurements during the period from July 2021 to April 2023 (Figure 11) shows lower percentage of cases within the requirements to that found for the comparisons with RadCalNet. In particular:

- CLMS TOC V1 shows typically around 15-25% of pixels within goal requirement (i.e., 3%) for most OLCI channels and typically 25-35% considering threshold level (5%). Significantly better results are found compared with ESA L2 SYN for all channels.
- The compliance for SLSTR channels is slightly lower to that found for equivalent OLCI channels. CLMS TOC V1 shows better compliance than ESA L2 SYN for VNIR channels (S1, S2 and S3) and same compliance for S5 channel.

3.1.2.1 Guarded acceptance

Figure 12 and Figure 13 show the CLMS TOC V1 guarded acceptance conformity testing (at coverage factor k = 2) results versus RadCalNet measurements in 2024 year for two OLCI (Oa07, Oa17) and SLSTR (S2, S5) channels, respectively. For sake of brevity the results for two channels are displayed in Figure 12 and Figure 13 (as most of them show similar results), but the summary for all channels is presented in Table 13 and Table 14. The uncertainties of both Sentinel-3 and RadCalNet measurements are considered for conformity testing. Once the apparent error and its expanded uncertainty are calculated, the conformity of each sample regarding goal and threshold requirements is determined (Camacho et al., 2024). It should be noted that this analysis cannot be performed for ESA L2 SYN since no uncertainty is associated with these products.

Main conclusions are:

- In case of OLCI, conformity testing reveals large uncertainties in the apparent error (Figure 12), with most cases as inconclusively conforming or non-conforming (>90%) and with no cases conclusively conforming (Table 13). Conclusively non-conforming pixels represent only around 5-10% and around 3-6% of the number of cases, depending on the goal and threshold requirement.
- Similar results are found for SLSTR S1, S2 and S3 channels (Figure 13, Table 14), with mostly inconclusively cases due to large uncertainties in the apparent error.
- In case of SLSTR S5 and S6 channels located in the SWIR, most cases are conclusively non-conform, explained in the large error due to the lack of ESA calibration correction factors in CLMS TOC V1 processing chain.

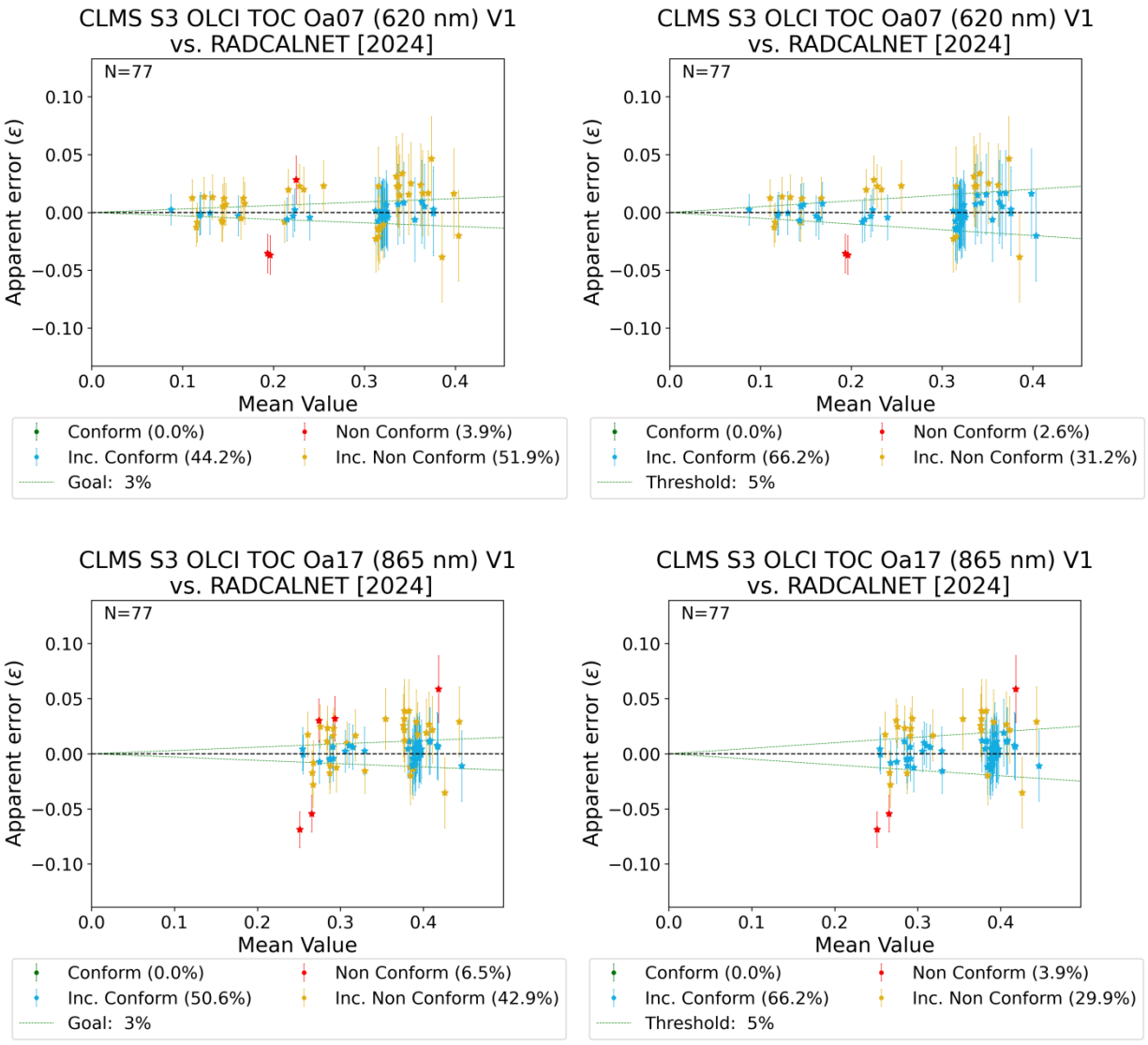


Figure 12: Conformity testing results (k=2) of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for OLCI Oa07 (top) and Oa17 (bottom) channels. Left and right side correspond to goal (3%) and threshold (5%) levels.

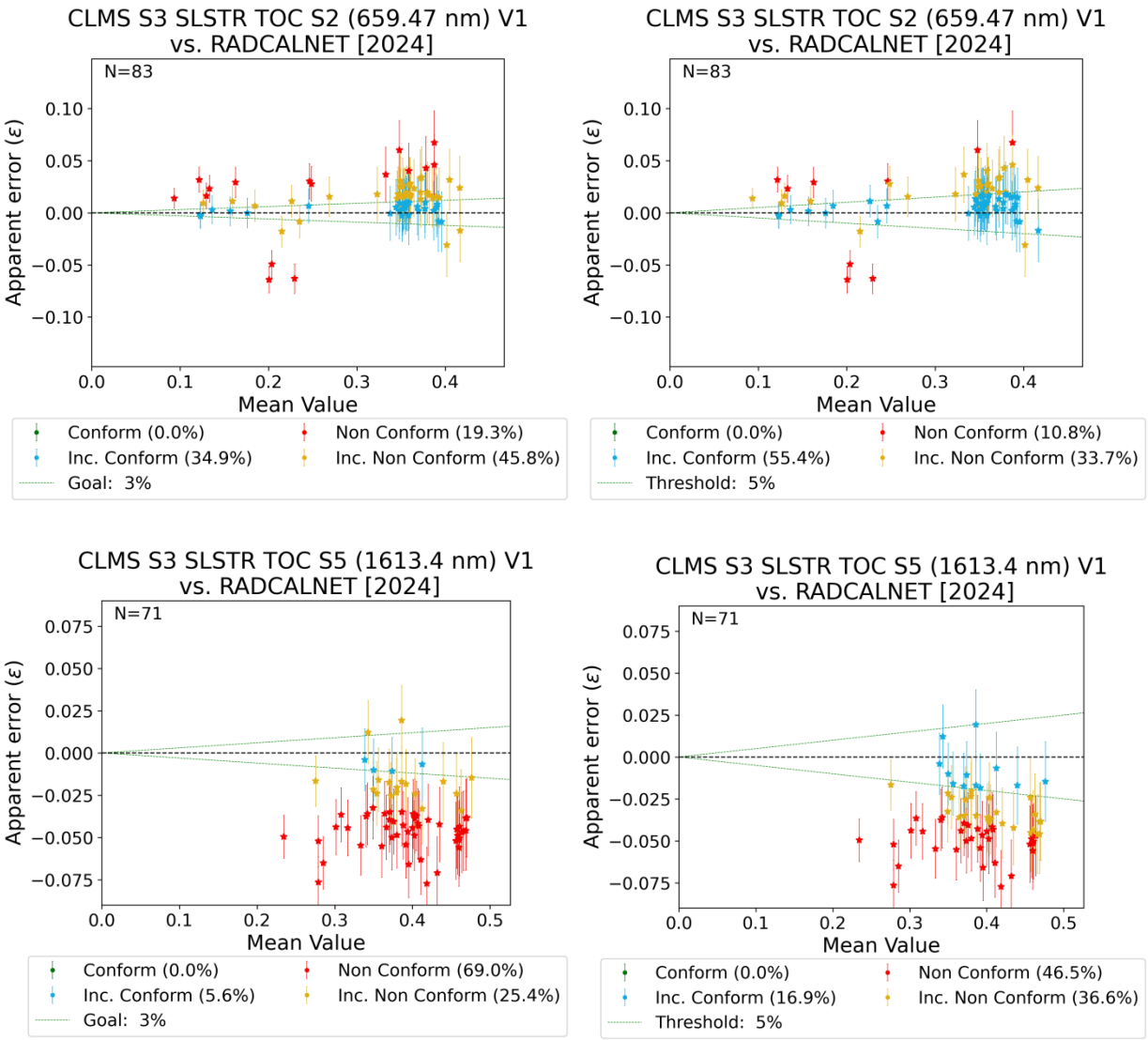


Figure 13: Conformity testing results (k=2) of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for SLSTR S2 (top) and S5 (bottom) channels. Left and right side correspond to goal (3%) and threshold (5%) levels.

Table 13: Conformity testing results of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for OLCI channels.

	% Goal (3%, k=2)				% Threshold (5%, k=2)			
OLCI Band	Conform	Inc. Conform	Inc. Non Conform	Non Conform	Conform	Inc. Conform	Inc. Non Conform	Non Conform
Oa02	0	11.7	70.1	18.2	0	28.6	53.2	18.2
Oa03	0	35.1	55.8	9.1	0	46.8	44.2	9.1
Oa04	0	44.2	46.8	9.1	0	62.3	29.9	7.8
Oa05	0	48.1	45.5	6.5	0	66.2	27.3	6.5
Oa06	0	45.5	46.8	7.8	0	67.5	27.3	5.2
Oa07	0	44.2	51.9	3.9	0	66.2	31.2	2.6
Oa08	0	46.8	58.1	5.2	0	63.6	32.5	3.9
Oa09	0	45.5	49.4	5.2	0	59.7	36.4	3.9
Oa10	0	39.0	55.8	5.2	0	57.1	40.3	2.6
Oa11	0	44.2	53.2	2.6	0	68.8	28.6	2.6
Oa12	0	46.8	46.8	6.5	0	64.9	32.5	2.6
Oa16	0	45.5	49.4	5.2	0	64.9	32.5	2.6
Oa17	0	50.6	42.9	6.5	0	66.2	29.9	3.9
Oa18	0	46.8	48.1	5.2	0	66.2	29.9	3.9
Oa21	0	29.9	58.4	11.7	0	55.8	37.7	6.5

Table 14: Conformity testing results of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (January 2024 – December 2024 period) for SLSTR channels.

	% Goal (3%, k=2)				% Threshold (3%, k=2)			
OLCI Band	Conform	Inc. Conform	Inc. Non Conform	Non Conform	Conform	Inc. Conform	Inc. Non Conform	Non Conform
S1	0	19.3	56.6	24.1	0	36.1	45.8	18.1
S2	0	34.9	45.8	19.3	0	55.4	33.7	10.8
S3	0	45.8	43.4	10.8	0	69.9	20.5	9.6
S5	0	5.6	25.6	69.0	0	16.9	36.6	46.5
S6	0	7.0	14.1	78.9	0	12.7	15.5	71.8

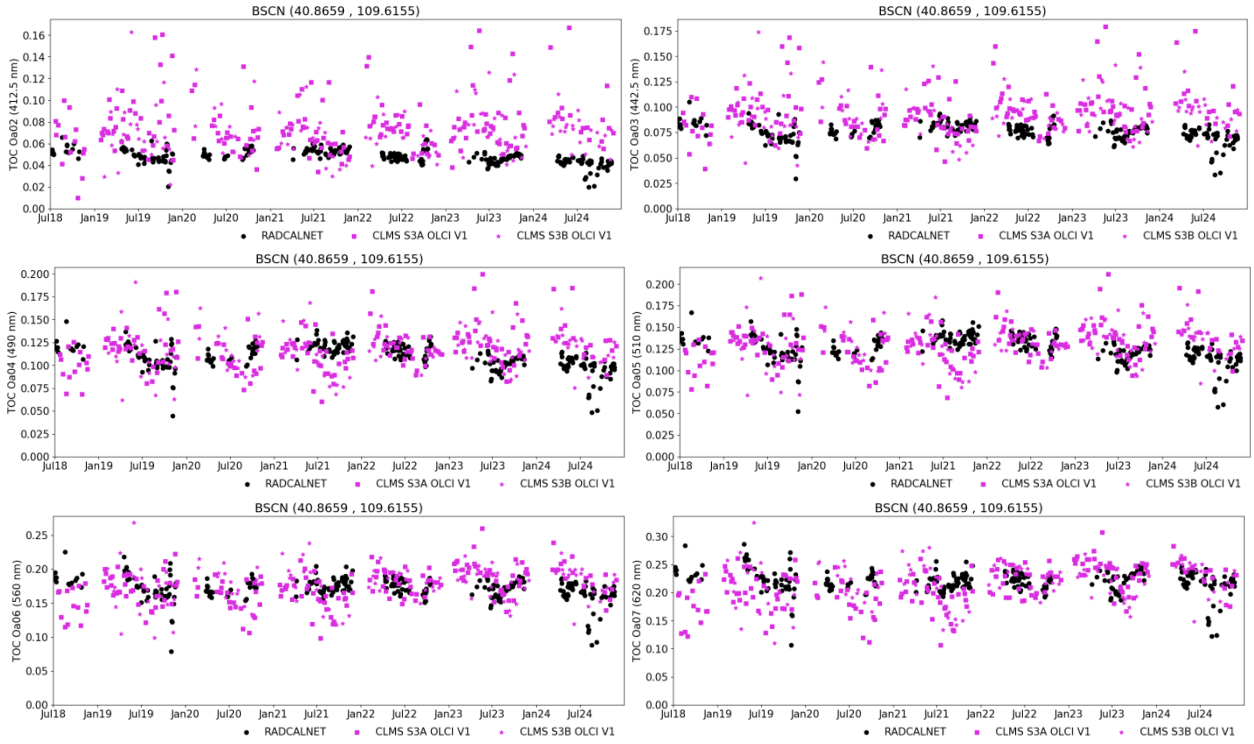
3.1.3 Temporal realism

In order to evaluate the temporal realism of CGLS Sentinel-3 TOC V1, temporal trajectories of all OLCI and SLSTR channels are displayed and qualitatively compared with RadCalNet surface reflectance measurements over the four sites during the period from July 2018 to December 2024, with focus on the recent 2024 year. Figure 14 (BSCN), Figure 16 (GONA), Figure 18 (LCFR) and Figure 20 (RVUS) show the temporal trajectories for OLCI channels whereas Figure 15 (BSCN),

Figure 17 (GONA), Figure 19 (LCFR) and Figure 21 (RVUS) show the temporal trends for SLSTR channels.

Main conclusions are:

- The temporal trajectories of surface reflectances of CGLS Sentinel-3 TOC V1 retrieved for both OLCI and SLSTR sensors show similar temporal trends than RadCalNet measurements, well capturing the seasonality observed over LCFR and the stable values of GONA. Slight temporal variability is found for RVUS and BSCN.
- The temporal trajectories for 2024 are consistent with previous years. Temporal trends between Sentinel-3A and B are also consistent.
- Additionally, the temporal trends over GONA pseudo-invariant site show:
    - For OLCI channels (Figure 16), positive differences (CLMS > RadCalNet) are clearly observed for the period from July 2018 to June 2022 (i.e., period affected by geolocation issue) and no systematic bias is found from July 2022 onwards (i.e., period where geolocation issue is corrected).
    - In case of SLSTR (Figure 17) VNIR channels (S1, S2, S3), CLMS typically shows higher reflectances than RadCalNet measurements, but the bias seems to be slightly lower from July 2022 onwards. No improvements in terms of bias are found for SLSTR SWIR channels from July 2022 onwards.



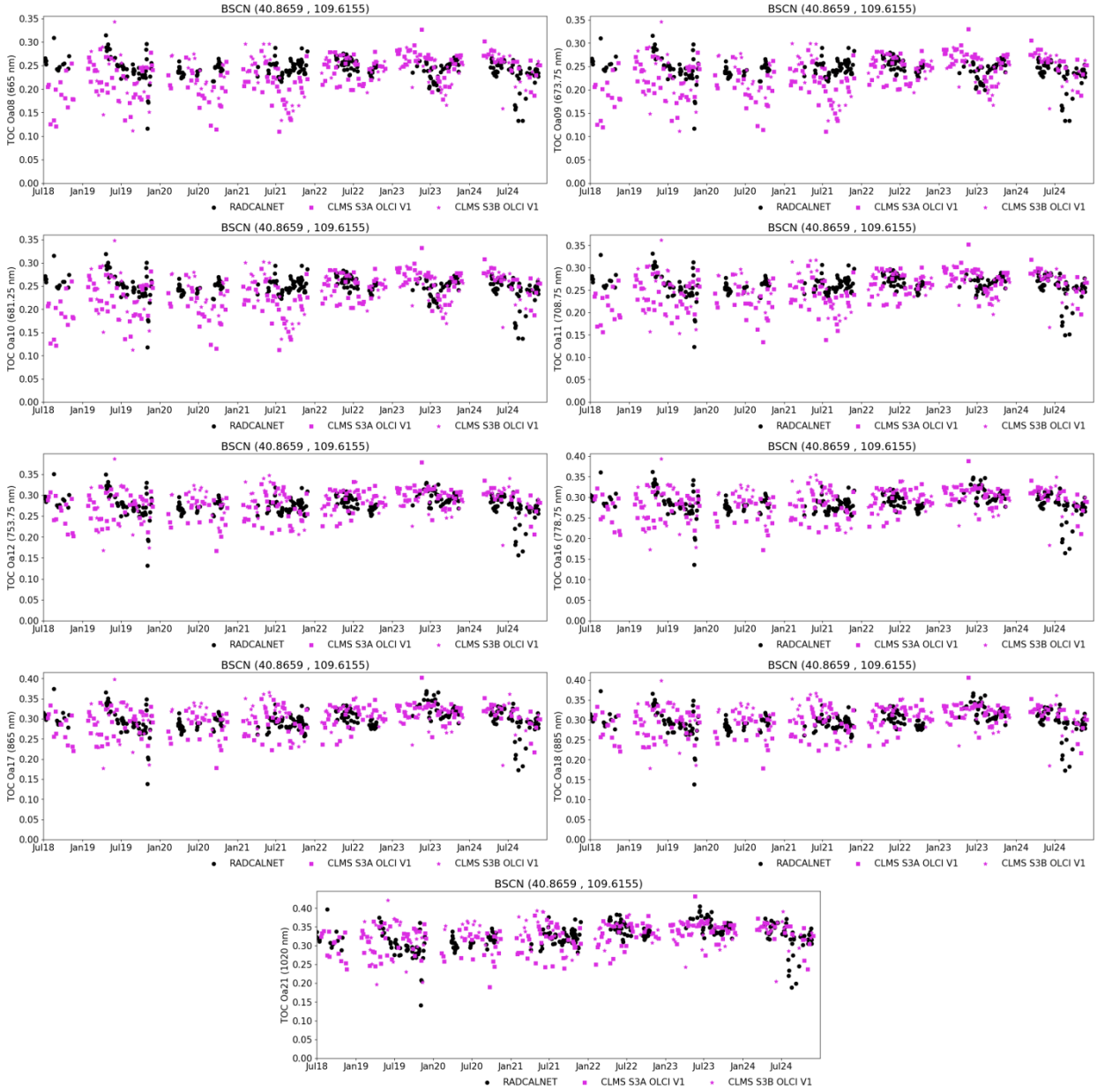


Figure 14: Temporal trajectories of CGLS Sentinel-3A (purple squares) and B (purple stars) TOC V1 (VZA<10º) for OLCI channels and RadCalNet measurements (black dots) for the period from July 2018 to December 2023 for BSCN site.

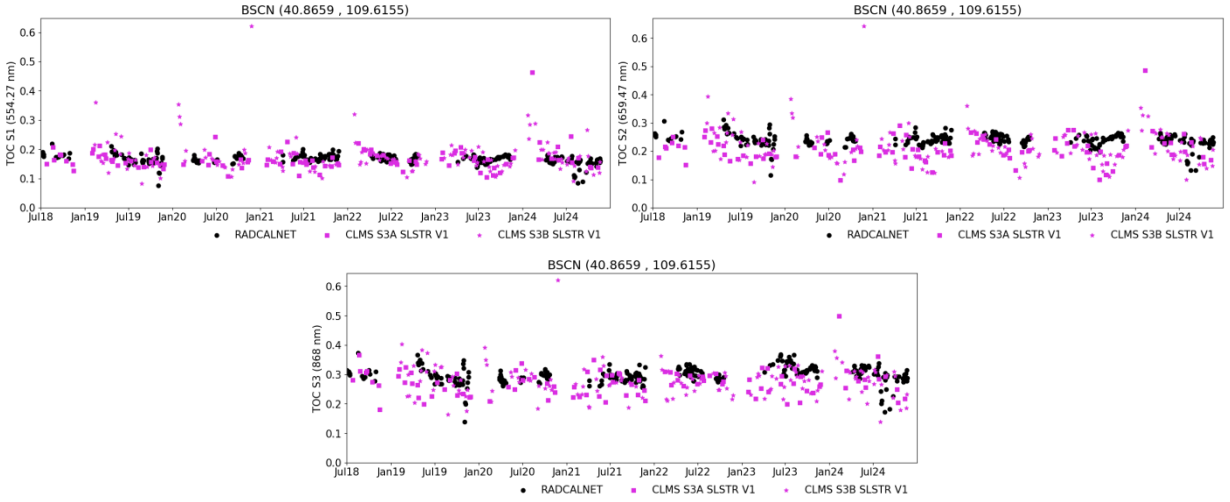
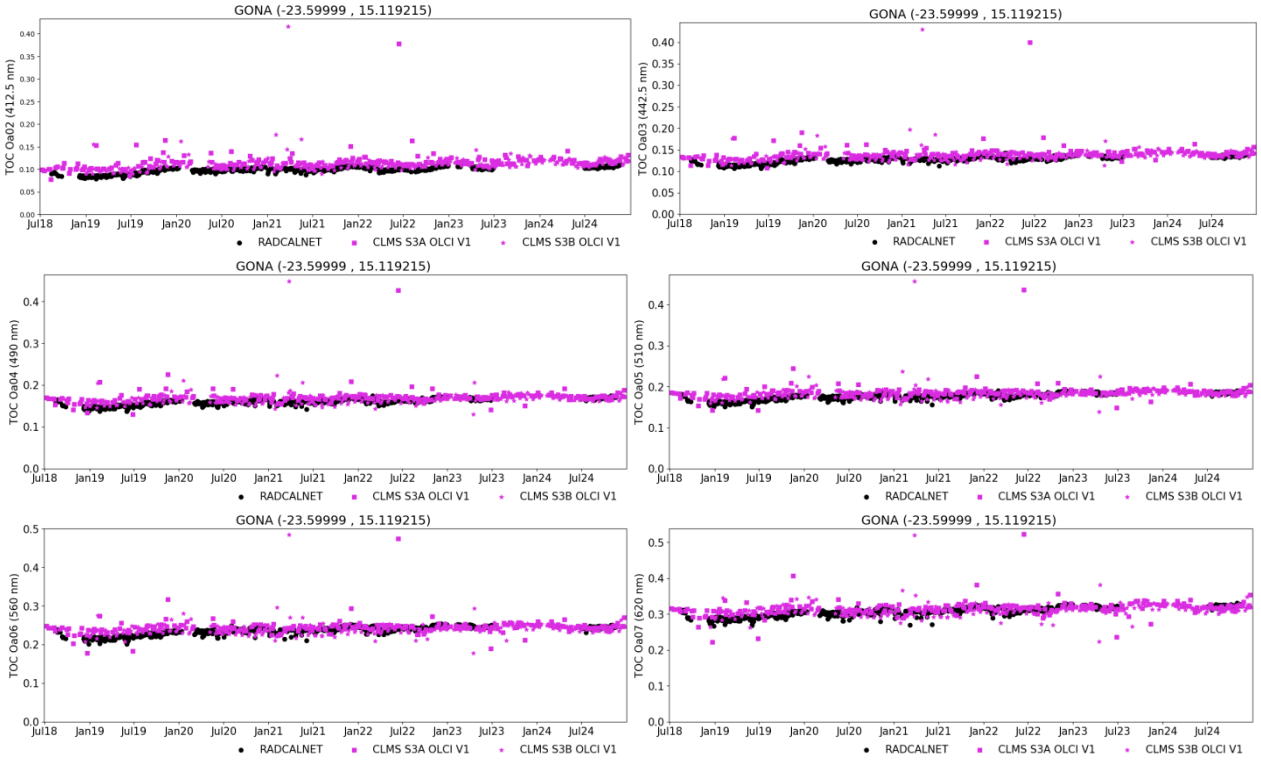


Figure 15: Temporal trajectories of CGLS Sentinel-3A (purple squares) and B (purple stars) TOC V1 (VZA<10º) for SLSTR channels and RadCalNet measurements (black dots) for the period from July 2018 to December 2023 for BSCN site.



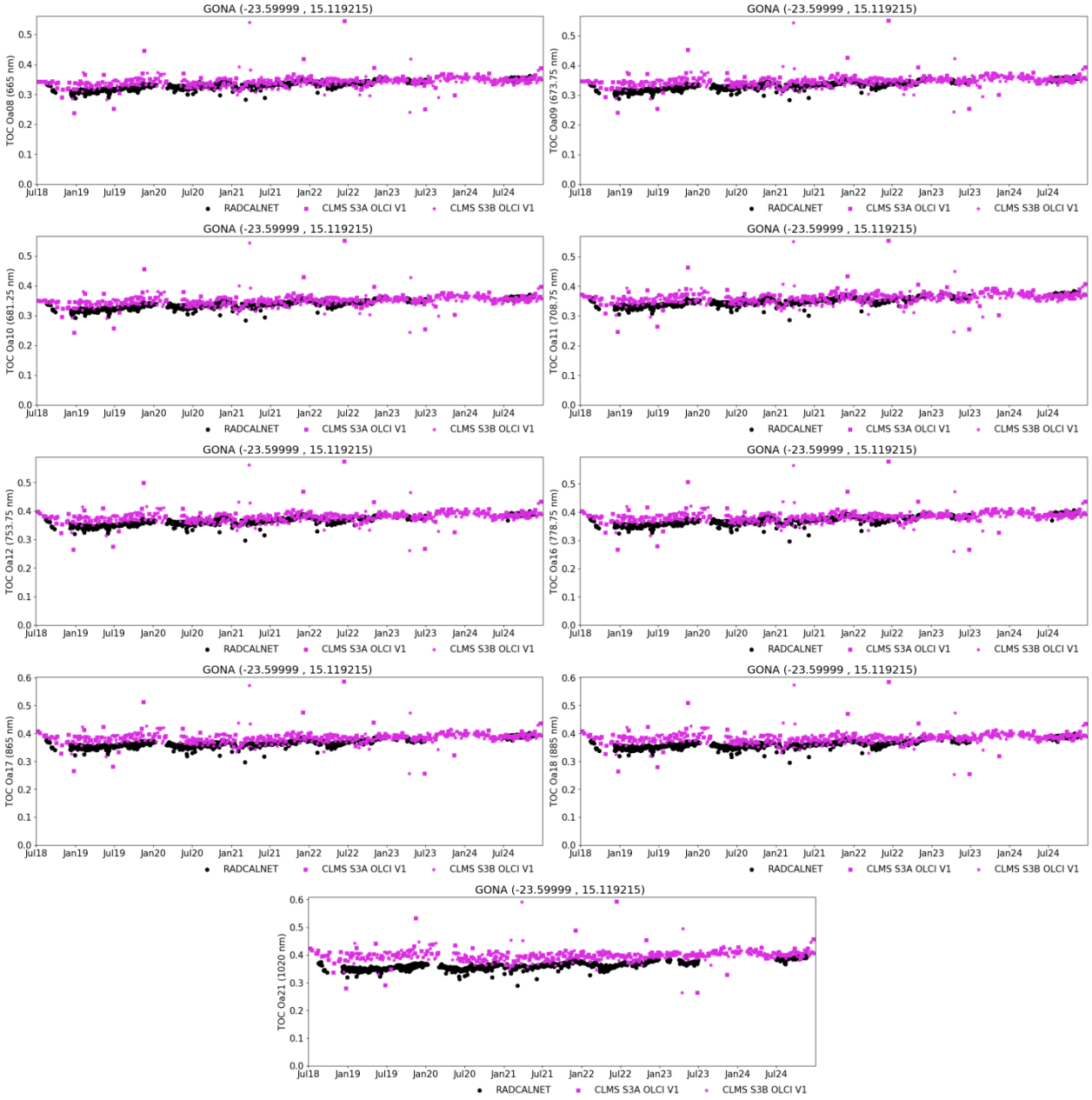


Figure 16: Same as in Figure 14 but for GONA site.

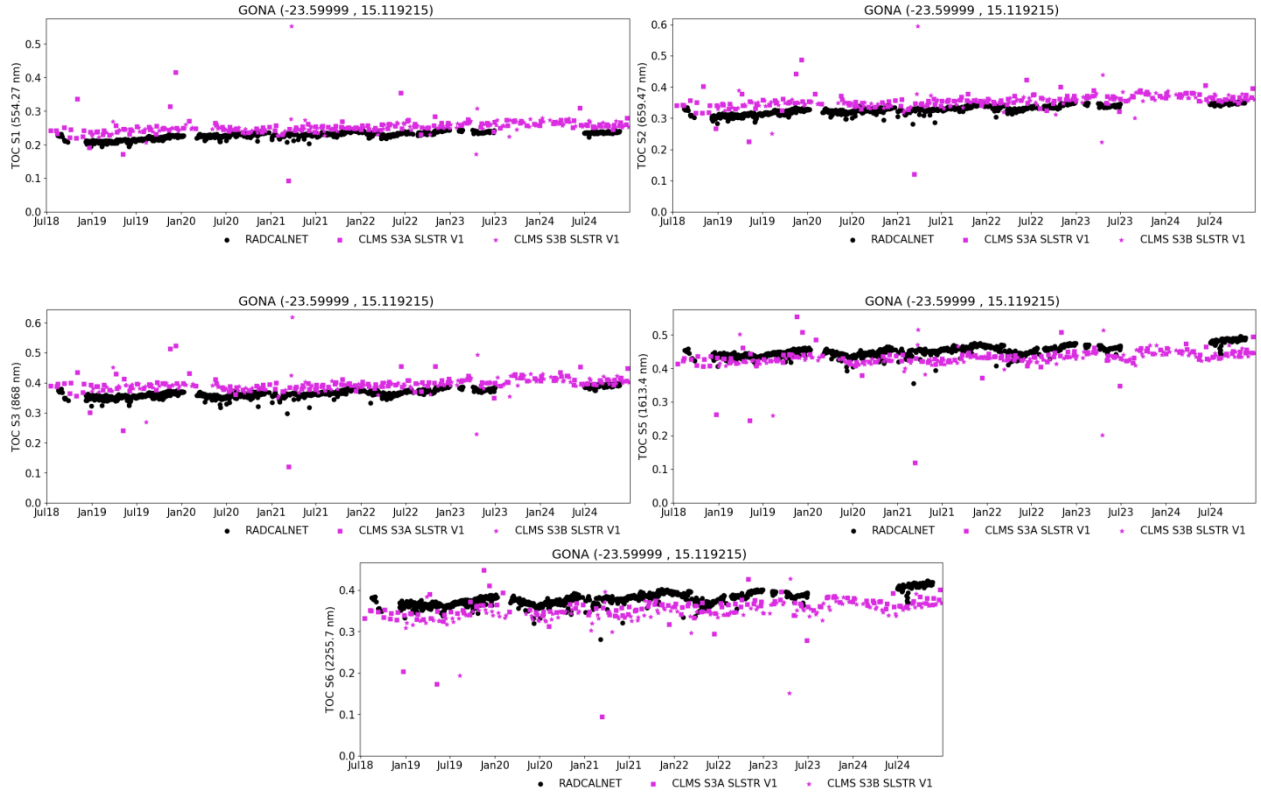
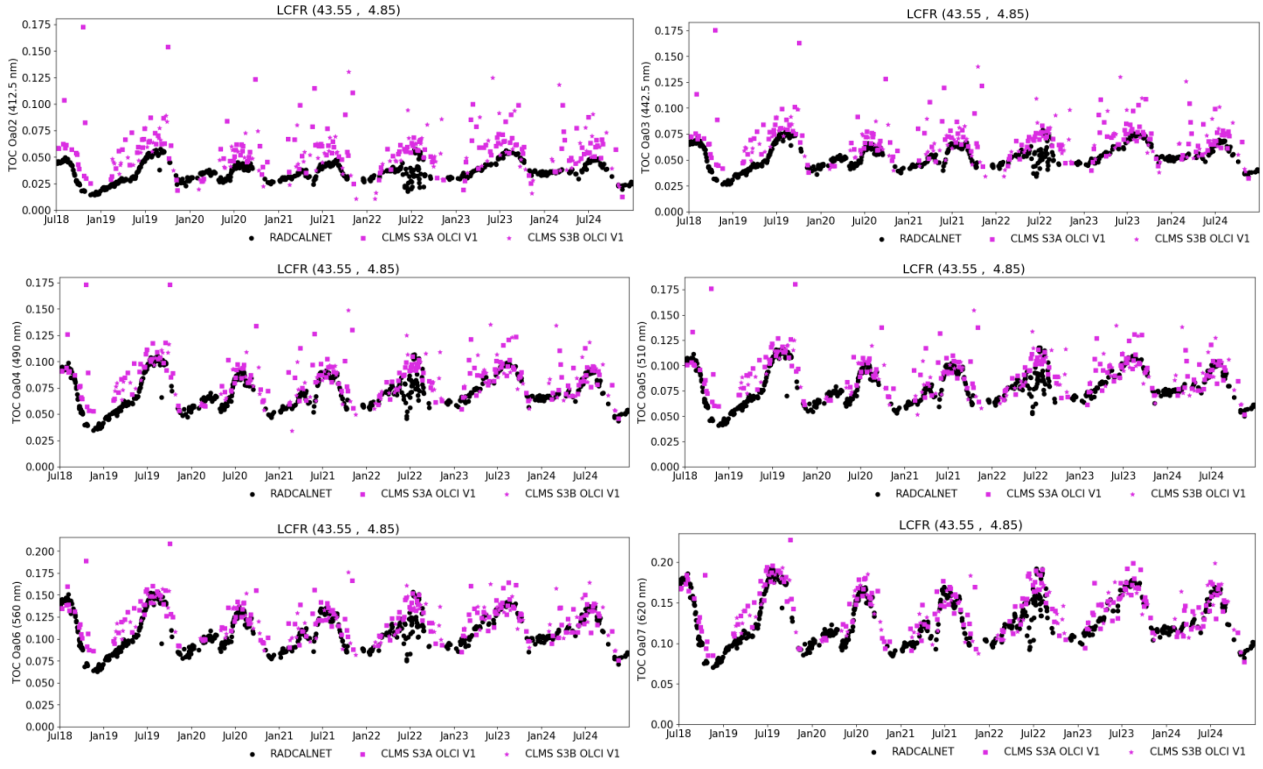


Figure 17: Same as in Figure 15 but for GONA site and SLSTR channels.



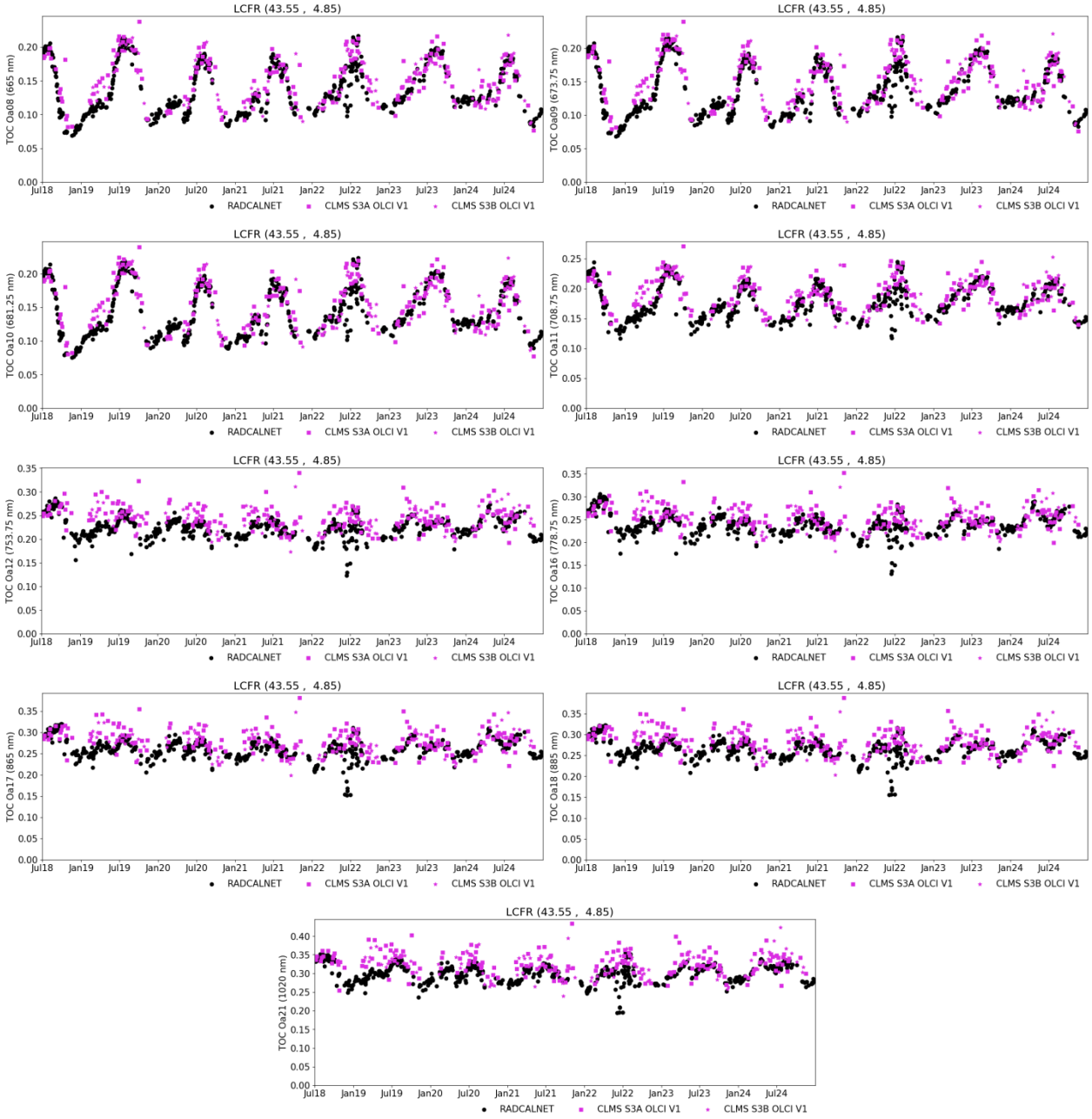


Figure 18: Same as in Figure 14 but for LCFR site.

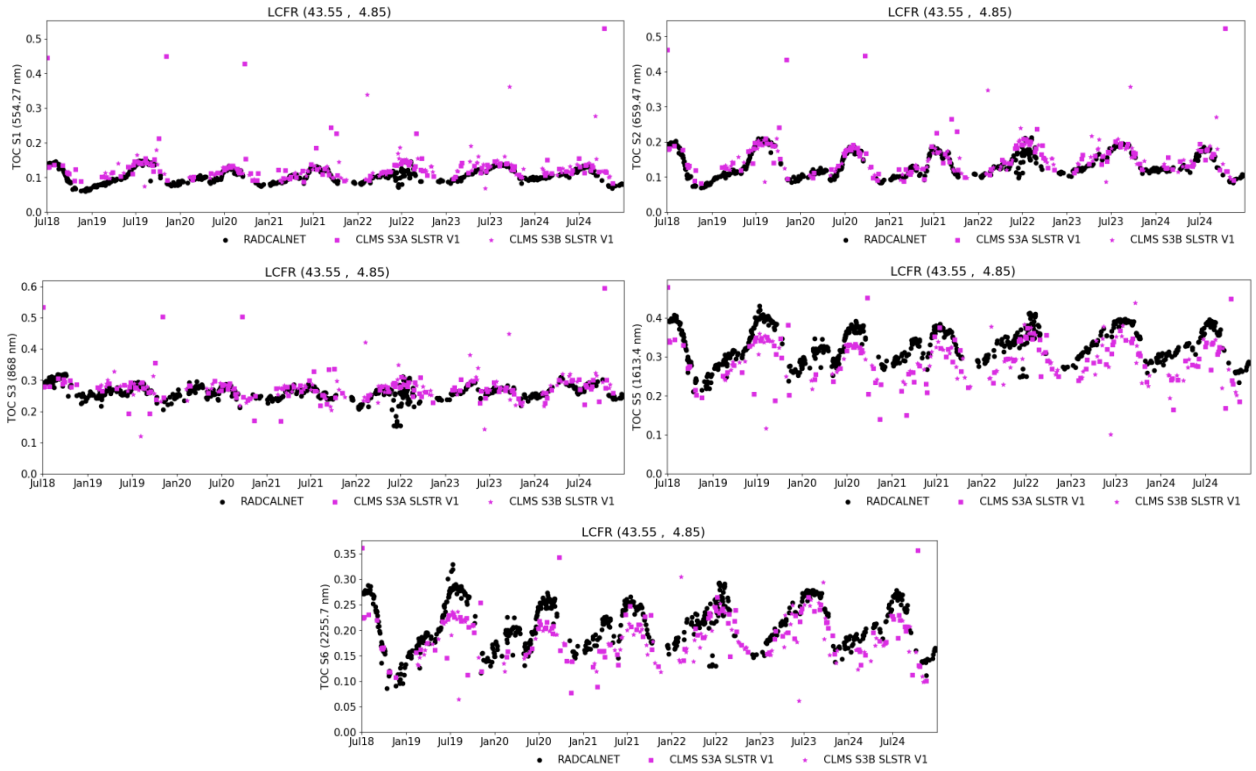
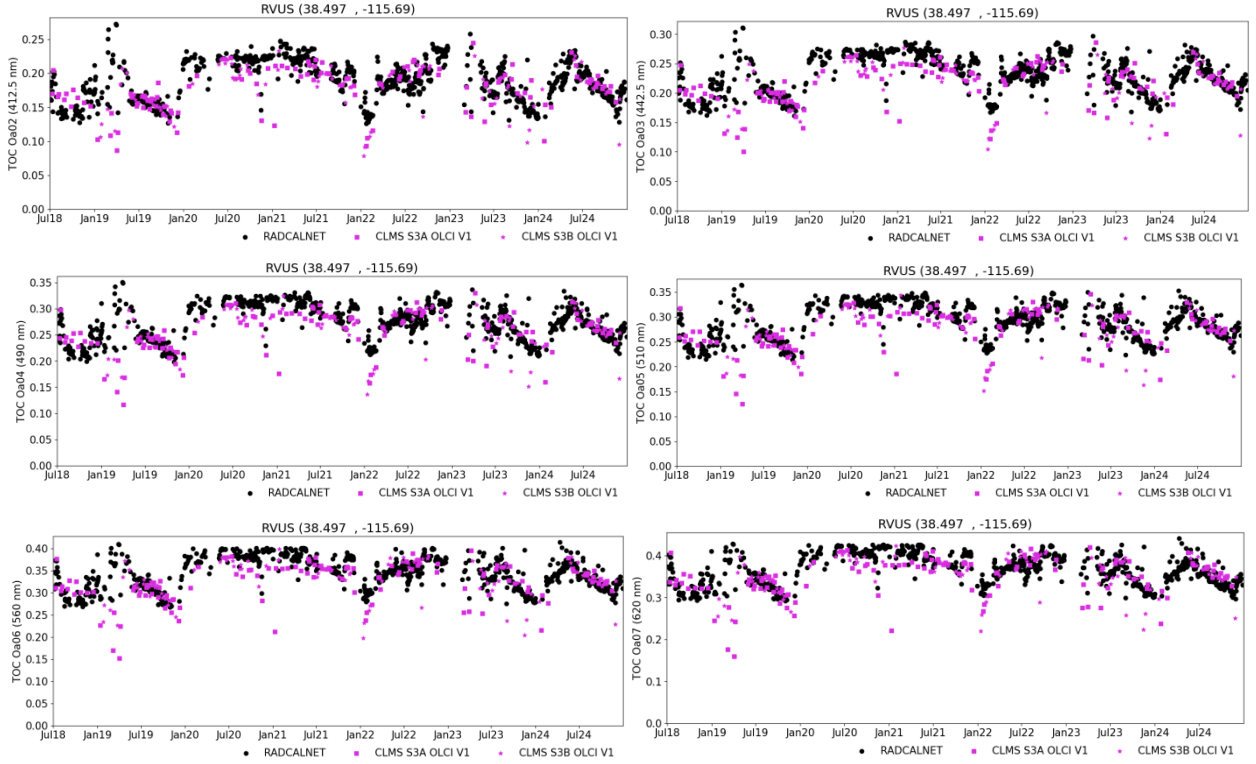


Figure 19: Same as in Figure 15 but for LCFR site.



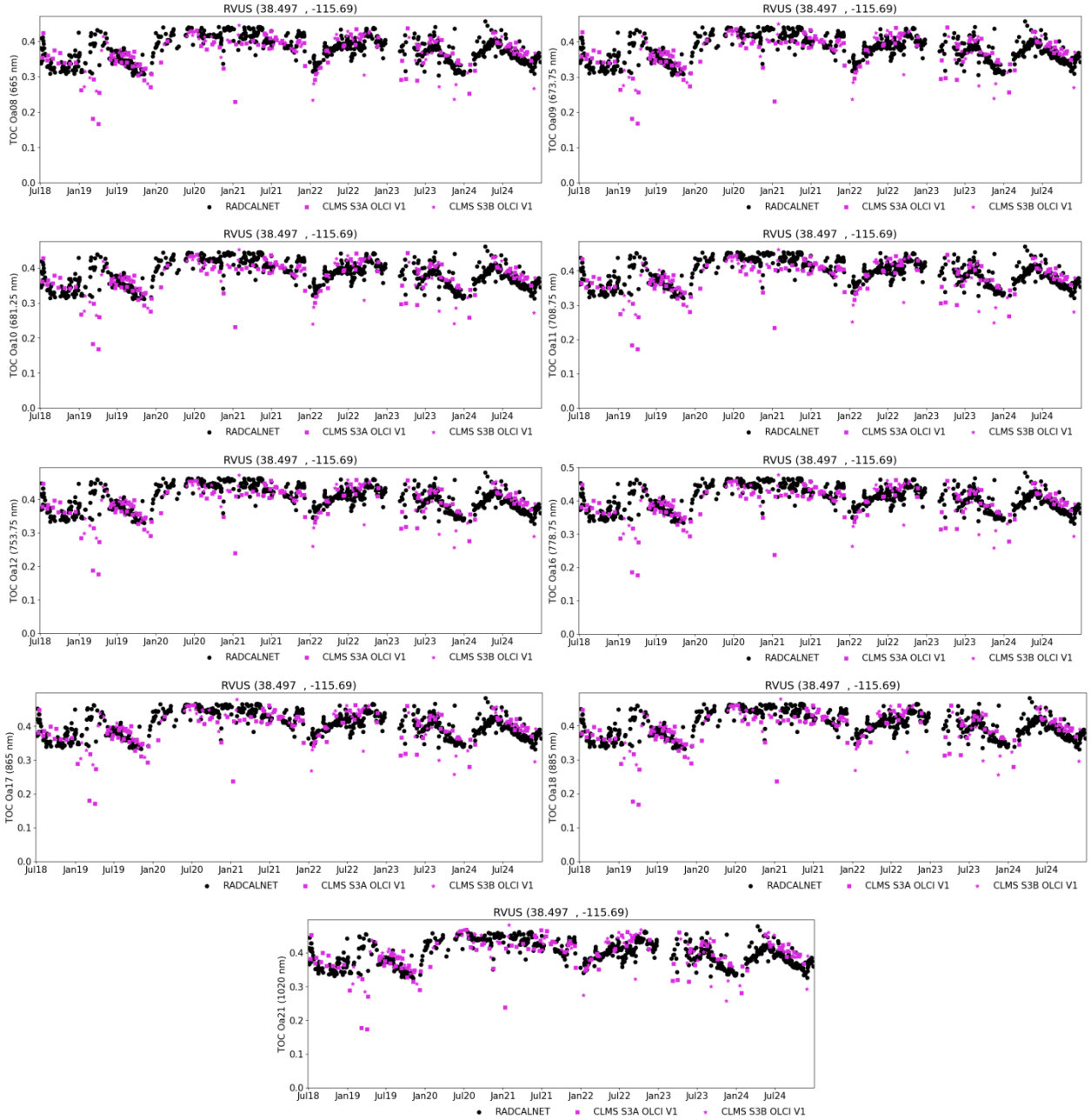


Figure 20: Same as in Figure 14 but for RVUS site.

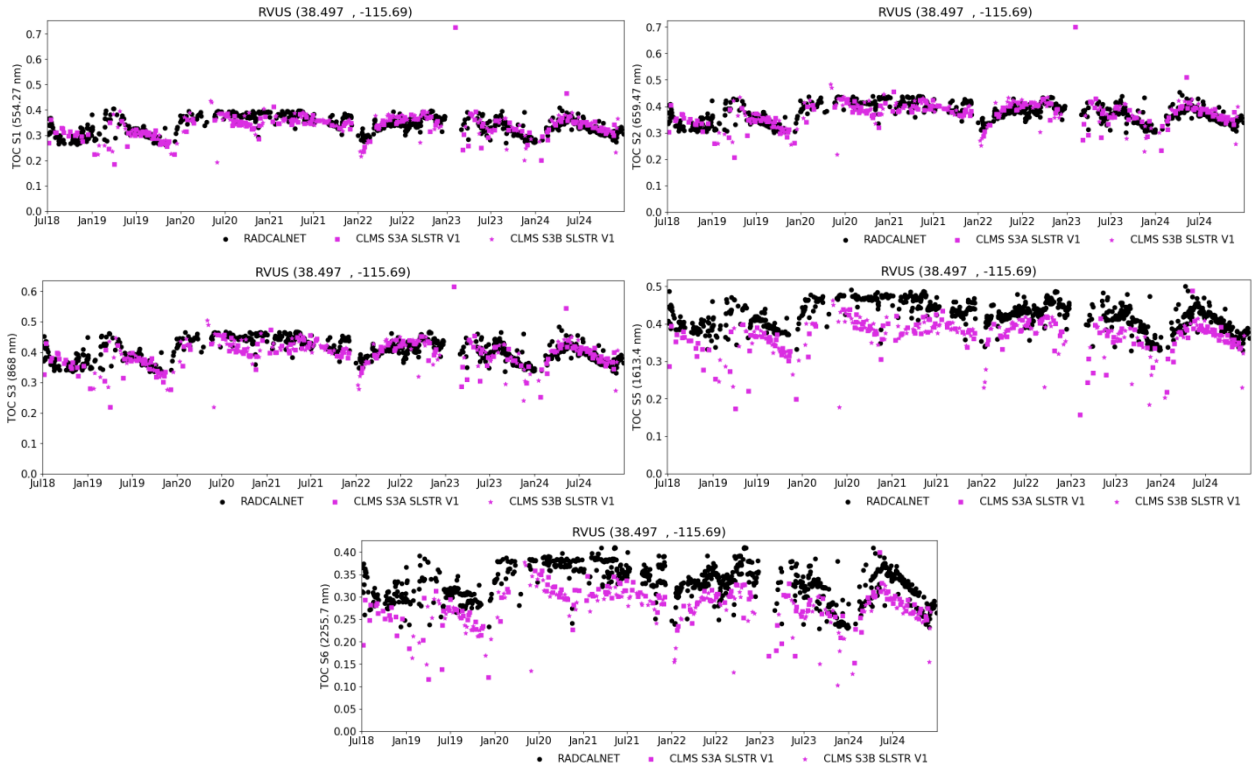


Figure 21: Same as in Figure 15 but for RVUS site.

4 CONCLUSIONS

The annual scientific validation of CLMS Sentinel-3 OLCI and SLSTR TOC reflectances V1 has been performed for recent 2024 year. CLMS TOC V1 products are evaluated against ground data from CEOS RadCalNet in 2024 and complemented with datasets of HYPERNETS (2021-2023 period due to their availability). ESA L2 SYN products are also compared with ground measurements (concomitant samples with CGLS product) for benchmarking.

The direct validation of TOC reflectances V1 retrievals close to nadir (VZA < 10º) compared with CEOS RadCalNet multi-temporal data from 4 sites (desert and sparse vegetation) shows similar results than previous periods and cumulative whole period (July 2018 – December 2024) in terms of accuracy (mean bias). However, better results are found for 2024 products compared with the whole cumulative and previous periods in terms of overall uncertainty (RMSD). Particularly, main conclusions for 2024 year are:

- CLMS TOC V1 shows good accuracy for OLCI channels, with mean bias and MD typically within ±2% for most channels [Oa05 – Oa18] and overall uncertainty (RMSD) typically better than 6%. The conformity testing (simple acceptance) shows that typically around 40-50% of pixels are within goal requirement (i.e., 3%) and 55-70% within threshold level (5%). CLMS TOC V1 shows significantly better results than ESA L2 SYN: B typically within 3% - 12%, RMSD within 7% - 15%, around 10-40% and 25-55% of pixels within goal and threshold.
- CLMS TOC V1 shows positive bias of 7%, 3% and 2% for VNIR S1, S2 and S3 SLSTR channels and negative bias of around -9% and -12% for S5 and S6 (as expected due to lack of calibration correction coefficients). Satisfactory conformity testing results (simple acceptance) are found for S2 and S3 channels, with around 40-45% of pixels within goal and 60-65% within threshold. On the contrary, lower compliance is found for S1, S5 and S6 (<12% goal and <25% threshold). CLMS TOC V1 shows better results than ESA L2 SYN for VNIR channels (B of 13.8%, 7.1% and 3.7% for S1, S2 and S3) and worse for SWIR S5 and S6 channels (bias of 6.8% and 7.7%) as expected due to incorporation of correction coefficients in the ESA L2 SYN processing.
- Conformity testing including satellite product and RadCalNet measurements uncertainties (i.e., guarded acceptance) provides mostly inconclusive results, with around half of the cases inconclusively non-conforming and the other half inconclusively conforming regarding the threshold level. These results are mainly because the uncertainties of the apparent error (i.e., satellite minus ground reference) are typically larger than user requirements. Only in the case of SLSTR S5 and S6, conclusively non-conform results are mainly obtained since calibration correction factor are not applied in CLMS TOC V1, and therefore large biases are found.
- The temporal trajectories of both CLMS TOC V1 from both OLCI and SLSTR sensors show similar temporal trends than RadCalNet measurements. Temporal trends are also consistent between different years and platforms (i.e., Sentinel-3A and -3B). Note that for GONA pseudo-invariant site, the positive systematic bias (CLMS > RadCalNet) observed for the

period from July 2018 to June 2022 is clearly reduced (almost no mean bias) from July 2022 onwards, explained to the correction of geolocation issue in the CLMS TOC V1 processing chain.

The direct validation of Sentinel-3 TOC reflectance V1 retrievals compared with HYPERNETS multi-temporal and multi-angular data from 3 sites (desert, agricultural and deciduous broadleaf forest) also shows satisfactory results: MD ranging from 4.4% up to 9.5% for OLCI channels and worse accuracy for SLSTR (MD of 14.3% and 9.4% for S1 and S2, MD of 3.6% for S3 MD of -10.3% for S5). ESA L2 SYN showed worse agreement from most APU metrics and channels.

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ANNEX I. RELEVANT STATISTICS (N, RELATIVE B AND RELATIVE RMSD) FOR CLMS TOC REFLECTANCES V1 OLCI VS. RADCALNET MEASUREMENTS FOR VARIOUS PERIODS

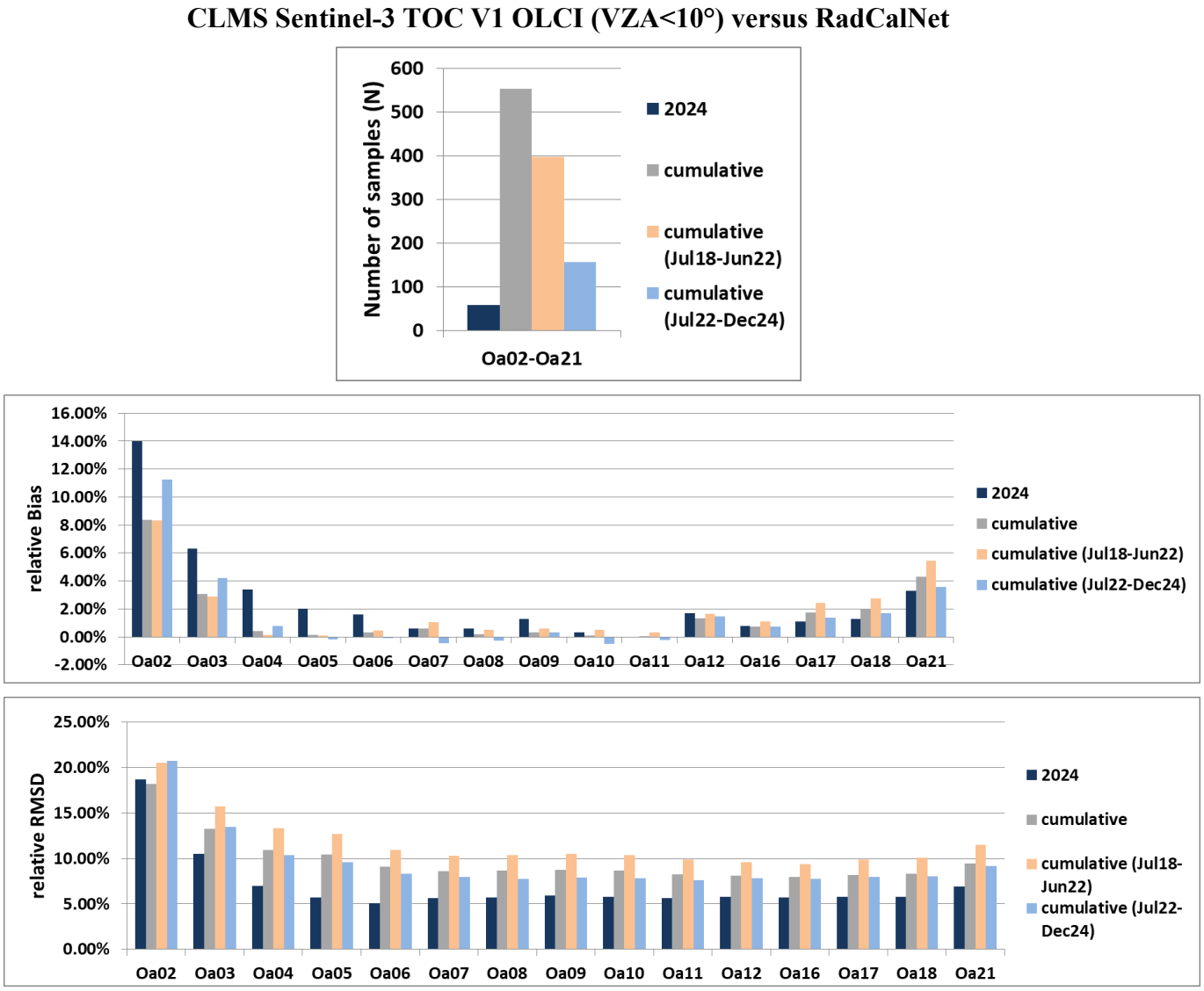


Figure 22: Number of samples (top), relative B (middle) and relative RMSD (bottom) of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (OLCI channels) for different periods: 2024 (dark blue), cumulative period from Jul'18 to Dec'24 (grey), cumulative period from Jul'18 to Jun'22 (orange, period with geolocation issue) and cumulative period from Jul'22 to Dec'24 (light blue, period without geolocation issue).

ANNEX II. RELEVANT STATISTICS (N, RELATIVE B AND RELATIVE RMSD) FOR CLMS TOC REFLECTANCES V1 SLSTR VS. RADCALNET MEASUREMENTS FOR VARIOUS PERIODS

CLMS Sentinel-3 TOC V1 SLSTR (VZA<10°) versus RadCalNet

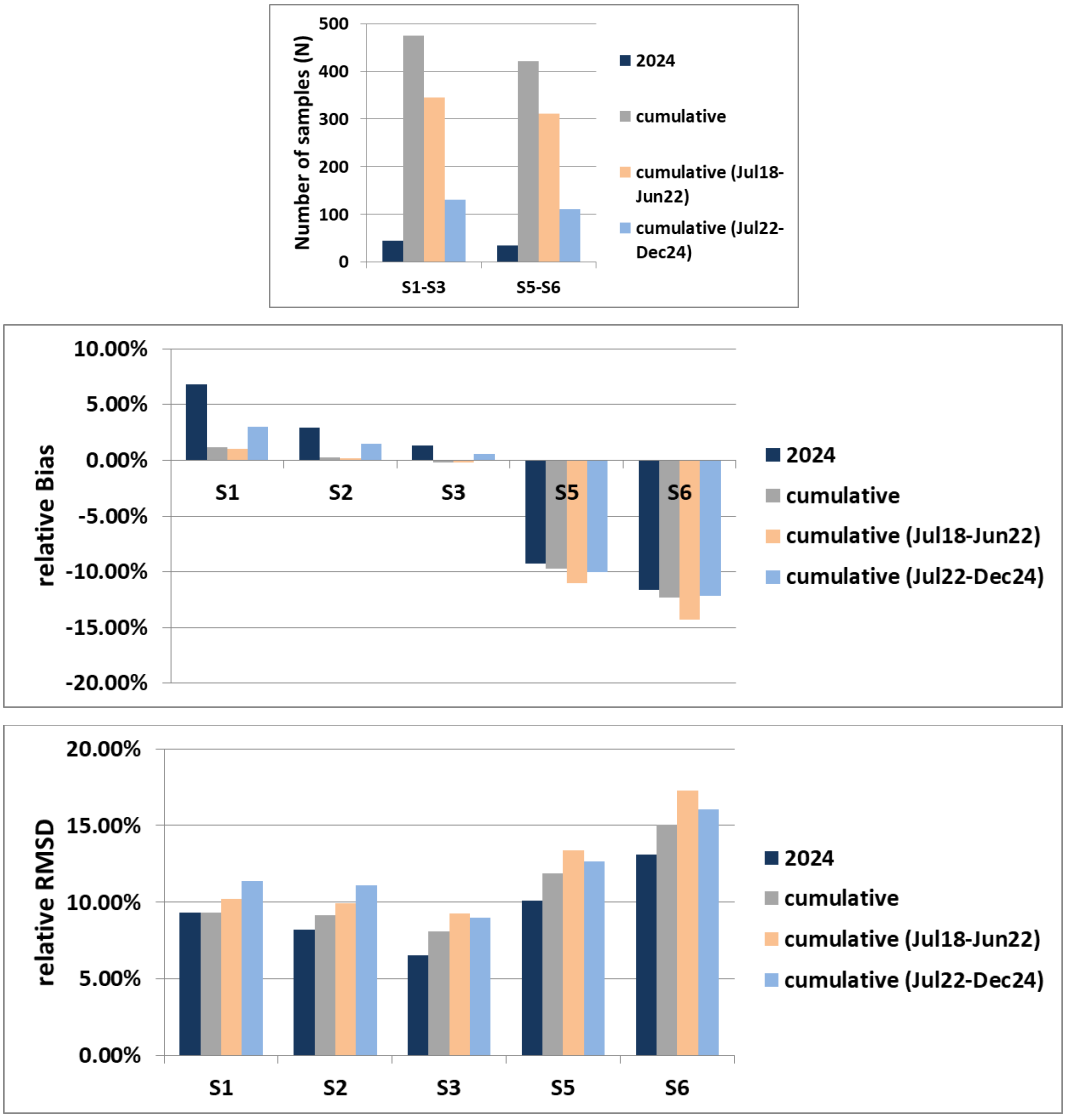


Figure 23: Number of samples (top), relative B (middle) and relative RMSD (bottom) of CLMS Sentinel-3 TOC V1 (VZA<10°) versus RadCalNet measurements (SLSTR channels) for different periods: 2024 (dark blue), cumulative period from Jul'18 to Dec'24 (grey), cumulative period from Jul'18 to Jun'22 (orange, period with geolocation issue) and cumulative period from Jul'22 to Dec'24 (light blue, period without geolocation issue).